Prabhāsa: Pāṇinian Structured Pretraining for Small Language Models

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Abstract:
Data-efficient pretraining of small language models remains an open problem, with limited evidence that linguistically grounded inductive biases improve downstream performance under fixed training budgets. This paper evaluates Prabhāsa, a controlled experimental framework examining whether Pāṇinian morphosyntactic structure, specifically through morpheme-boundary embeddings and role-aware masking strategies, yields measurable gains on the BabyLM official suite across 100M and 10M parameter models. Experiments employ a pre-registered hypothesis hierarchy, testing both masking-distribution mechanisms and auxiliary objective supervision with matched baselines, three or more seeds per arm, and paired bootstrap inference to control family-wise error. Three findings emerge. First, architecture (RoPE positional encoding) and objective choice (pure MLM without CLM) drive scale-dependent improvements: the 100M model achieves 73.06 BLiMP and 55.99 text-average, compared to the Strict-track baseline of 74.53 BLiMP, while the 10M hybrid model reaches 64.09 BLiMP compared to the Strict-Small baseline of 65.08 BLiMP. Second, kāraka role-stratified masking shows no significant causal effect at matched budget (F2 produces delta +0.10 95% CI [-0.99, +1.20]). Third, a supervised kāraka-role-prediction auxiliary objective shows no statistically significant effect across five seeds (delta +0.76 95% CI [-0.74, +2.27]). The contribution of the Pāṇinian framing lies in transparent mechanism design and interpretability rather than empirical lift on this benchmark. These results are reproducible, openly released on HuggingFace and GitHub, and provide a rigorous baseline for future work on linguistically motivated pretraining.

Keywords: language models; data-efficient pretraining; BabyLM; Pāṇinian grammar; morphology-aware masking; Navya-Nyāya; masked language modelling

1 Introduction

The scaling of language model pretraining has driven rapid progress in natural language understanding and downstream task performance. Yet the relationship between data efficiency, inductive bias, and model behaviour remains underspecified, particularly at smaller scales. Small language models operating under strict computational budgets (10M to 150M parameters) offer a controlled experimental testbed for understanding which architectural and training choices yield measurable improvements, free from the confounds that emerge when training frontier-scale models. Understanding what matters at small scales is therefore both scientifically and practically important, as it constrains the design space for data-efficient pretraining and clarifies which inductive biases translate across scales.

One broad class of inductive biases derives from linguistic theory. Classical Pāṇinian grammar, developed over more than a millennium [1], encodes compositional structure at the morpheme level through morphological segmentation and inflectional rules, and at the argument level through systematic kāraka categories that denote semantic roles. These categories and their combinatorial rules are known to capture syntactic and semantic patterns in natural language. Whether embedding such structure as explicit inductive biases in the pretraining procedure improves data efficiency for small language models remains an open empirical question. The motivation is intuitive: if a linguistically grounded prior reduces the effective hypothesis space the model must search, it should require fewer training tokens to achieve a given level of generalisation.

This paper investigates this hypothesis through a controlled experimental study of Pāṇinian-motivated mechanisms within the BabyLM constrained-data setting [2]. The program, called Prabhāsa, comprises three research hypotheses that structure the empirical inquiry.

H1 (the primary hypothesis) asks whether morpheme-boundary structure, implemented as Vidyut N-hot embeddings [3] combined with role-aware masking (kāraka-stratified masking probability), measurably improves compositional generalisation on English syntax benchmarks (BLiMP agreement and argument-structure subsets) under matched training budgets and controlled comparisons. This hypothesis decomposes into two mechanistically distinct sub-questions that isolate different potential levers.

RQ-A (masking causality) asks whether at matched mask-rate budget, kāraka role-stratification of token masking carries causal efficacy on the targeted BLiMP paradigms. This question asks whether altering which tokens are masked, based on their morphological or semantic-role structure, affects downstream syntax performance.

RQ-B (auxiliary objective) asks whether a supervised auxiliary objective (token-level kāraka-role prediction from the contextualized embeddings of a pure-MLM model) improves BLiMP performance in cases where the masking lever from RQ-A was ineffective. This question asks whether explicit role supervision provides a signal the masking strategy alone does not.

H2 (synergy in scope) asks whether on the best H1-validated arm, a 6-phase Pañcāvayava fine-tuning scaffold reduces fallacious-inference rates and improves sample efficiency on reasoning-quality readouts versus a matched generic baseline.

H3 (epistemic constraint, out of scope) asks whether a GBNF schema coupled with a Z3 vyāpti verifier can enforce causal-validity constraints by construction.

The primary evaluation focuses on H1 with pre-registered thresholds such as at least +1.0 percentage point gain on targeted paradigm subsets, with significance assessed at at least 3 independent seeds using paired bootstrap resampling and Holm-Bonferroni family correction. Pre-registration serves a critical role: it prevents post-hoc rationalisation of weak or null findings and enforces statistical discipline before results are known.

This work contributes to the empirical understanding of linguistic inductive biases in small-model pretraining in several ways. First, RQ-A and RQ-B are tested independently using budget-matched arms to isolate the causal role of masking distribution from that of auxiliary supervision. This separation rules out confounds (such as differences in effective mask rate) that have plagued earlier single-arm comparisons and prevented clear mechanistic conclusions. Second, an orthogonal finding during the ablation revealed that the choice between pure masked-language modelling (MLM) and hybrid MLM plus causal-language-modelling (CLM) objectives is scale-dependent. At 10M tokens, the two objectives are statistically equivalent (pure-MLM 63.58 percent plus or minus 1.73 percent versus hybrid 64.09 percent plus or minus 0.26 percent on BLiMP, overlapping confidence intervals). At 100M tokens, pure-MLM substantially outperforms hybrid MLM plus CLM (73.06 percent versus 67.57 percent, a 5.49 percentage-point gain). This finding was validated independently and suggests that the CLM head's dilution of MLM training quality is a scale-dependent bottleneck in small-model pretraining. Third, all hypothesis tests employ at least 3 independent random seeds, paired bootstrap resampling within pre-registered target sets, and explicit pre-registration of outcome thresholds and inference procedures. A separate Tarka memo, representing the strongest objection to one's own finding, is prepared for each result to identify confounds and alternative explanations before closure. Fourth, code, data, model checkpoints, and per-seed results are released openly via HuggingFace model cards and the GitHub repository with full hyperparameter disclosure and configuration hashing, enabling replication and downstream use by the community.

To the best of the authors' knowledge, this is the first work to apply pre-registered, multi-seed, mechanistically-decomposed testing of Pāṇinian linguistic structure as an inductive bias for masked-language-model pretraining under strict data budgets. Prior work on linguistically motivated pretraining, including morpheme-aware objectives [4] and morphologically grounded word representations [5], has tested linguistic priors in isolation or on larger models. Prabhāsa combines independent ablation of two mechanistically distinct levers (masking versus auxiliary objective), multi-seed validation, pre-registered significance thresholds, and explicit control for confounds. The distinctive contribution is therefore the mechanistic decomposition and statistical discipline, rather than any individual technique; together, these elements constitute a novel experimental design for isolating the causal efficacy of linguistically motivated structure in small-model pretraining.

The empirical findings from H1 indicate that architectural choices (RoPE positional encoding [6]) and objective design (pure MLM without auxiliary CLM) are the primary levers for improving compositional generalisation at this scale (10M to 100M tokens). The Pāṇinian mechanisms, including masking stratification and auxiliary kāraka-role prediction, show null or marginal effects when tested with appropriate controls. Specifically, RQ-A (masking causality) yields a pre-registered null finding at 100M (Arm K versus Arm C: delta K minus C equals +0.10 percentage points, 95 percent confidence interval [negative 0.99, +1.20], non-significant). RQ-B (auxiliary objective) exhibits a weak, non-significant effect at 10M when tested across 5 seeds (delta Aux minus Base equals +0.76 percentage points, 95 percent confidence interval [negative 0.74, +2.27], t equals 1.41, df equals 4, not significant), with the single-seed positive result shrinking after adding more seeds, indicating seed variance rather than robust signal.

This outcome is scientifically valuable as it constrains the scope of claims about structure-driven pretraining, identifies the true levers (architecture and objective), and provides a rigorous baseline for null findings in the community. A well-documented null is a valid and necessary contribution, particularly when it overturns optimistic prior expectations and is grounded in pre-registered, multi-seed validation.

The remainder of this paper is structured as follows. Section 2 reviews related work on inductive biases in pretraining, data-efficient language model training, and computational treatments of Pāṇinian grammar and Navya-Nyāya. Section 3 describes the corpora, the synthetic Pāṇinian pre-pretraining dose, and the four dose arms. Section 4 introduces the model architecture and the Pāṇinian mechanisms, and states the research questions and their pre-registered thresholds. Section 5 describes the training stages, the fine-tuning procedure, and the evaluation protocol. Section 6 reports results for each research question with effect estimates and confidence intervals. Section 7 discusses the interpretation of the findings, their implications for small-model pretraining, and threats to validity. Section 8 concludes and outlines future work. Supplementary configuration details, per-seed tables, and reproducibility information are provided in Appendix A.

2 Related Work

Prabhāsa builds on a long tradition of research into data-efficient pretraining, morphologically-informed linguistic structure, and the role of formal grammatical frameworks in neural language modelling. This section situates the work within four overlapping research areas: the BabyLM controlled pretraining paradigm, morphology-aware masking strategies and curriculum learning, the computational treatment of Sanskrit and the Pāṇinian grammar, and the application of classical Indian logic (Navya-Nyāya) to reasoning and semantic inference. We close by identifying the novel synthesis that distinguishes the present approach.

The conventional scaling paradigm in language model pretraining prioritises large training budgets and extensive data corpora [7]. Recent work has reconsidered whether smaller models trained on limited compute can achieve reliable linguistic and structural competence through careful architectural and training choices. The BabyLM Challenge, introduced by Warstadt et al. [2], establishes a controlled benchmark with a fixed 10M-token training budget and evaluates models on compositional generalisation tasks (SCAN, COGS, CFQ, BLiMP) and standard NLP benchmarks (GLUE). This framing isolates the contribution of training methodology from dataset scale and permits direct measurement of data efficiency. The 2024 iteration, reported by Hu et al. [8], evaluates competing approaches on the official leaderboard suite, including new subword tokenisation strategies and masking regimes.

Several noteworthy submissions to the BabyLM challenge demonstrate that structural priors can improve pretraining efficiency. LTG-BERT [9] employs language-theoretic linguistic structure combined with subword boundary masking. ELC-BERT [10] integrates explicit morphological parsing to inform masking. GPT-BERT [11] adapts causal language modelling objectives to the BabyLM regime. These submissions recover small but consistent gains on compositional tasks by encoding structural priors in the masking schedule or tokenisation. The official BabyLM baselines (Strict BLiMP 74.53 and Strict-Small BLiMP 65.08) establish reference points for competitive submissions.

2.1 Morphology-Aware Pretraining and Curriculum Masking

The insight that masking schedules can encode linguistic structure motivates recent work on morphologically-informed adaptive masking. Traditional masked language modelling (MLM) [12] masks input tokens uniformly at random, treating morphological boundaries and syntactic constituencies as irrelevant. Morpheme-aware masking strategies, by contrast, condition masking probability on linguistic structure, thereby increasing the difficulty of learning tasks that explicitly require compositional reasoning. Curriculum masking, in which masking difficulty escalates across training, has shown benefits for data-efficient pretraining. Søgaard et al. [13] demonstrate that curricula based on linguistic complexity can accelerate convergence. More recent work on adaptive masking schedules, as applied in several BabyLM submissions, dynamically adjusts masking probability based on token properties such as morpheme boundaries, argument structure, or dependency-parse depth. This approach trades increased computational cost during preprocessing against potential gains in sample efficiency and final model capacity on structured tasks.

Sanskrit and morphologically complex languages present natural testbeds for morpheme-aware pretraining. The Vidyut toolkit [3] provides morphological analysis and segmentation for Sanskrit. The Digital Corpus of Sanskrit [14] and the GRETIL collection [15] supply large Sanskrit text corpora. Huet's Sanskrit Heritage platform [16] includes morphological generation and parsing. These resources enable direct measurement of morphological competence, including sandhi restoration, case-marking agreement, and stem-inflection pairing, on low-resource non-English linguistic systems where the utility of morphological structure is empirically unambiguous.

2.2 Subword and Morpheme Tokenisation

The choice of tokenisation scheme has profound implications for pretraining efficiency. Byte-pair encoding (BPE) [17] and WordPiece are standard, but neither encodes morphological boundaries explicitly. Morpheme-aware tokenisation, in which vocabularies are constrained to true morphemes or morpheme-like units, reduces vocabulary size and enforces that learned representations reflect linguistic structure.

Botha and Blunsom [5] train segmentation models to recover morphemic boundaries and demonstrate downstream benefits for morphologically complex languages. More recently, Yaltik et al. [4] show that morpheme-based segmentation in multilingual NMT systems improves low-resource language performance. For Sanskrit, morpheme tokenisation is especially natural: the Pāṇinian grammar (circa 500 BCE) defines the language via morphological rules, and Vidyut implements this classical framework computationally [3]. Tokenising Sanskrit into morphemes directly instantiates this grammar.

2.3 Rotary Embeddings and Modern Positional Encodings

Transformer models require explicit position signals. Vaswani et al.'s sinusoidal embeddings [18] were standard in early work. More recent innovations include relative positional biases [19], ALiBi [20], and Rotary Position Embeddings (RoPE) [6]. RoPE applies complex-number rotations in the embedding space, encoding position information in a way that respects the rotational structure of attention. RoPE has become standard in recent large models and provides strong compositional generalisation on length extrapolation tasks. The present work employs RoPE as a baseline positional encoding and evaluates whether morphological structure provides orthogonal benefits.

2.4 Optimiser Choice and the Muon Optimizer

The optimiser used during pretraining influences both convergence speed and final model capacity. Standard choices include Adam [21] and AdamW [22]. Recent work has revisited the role of the optimiser in small-model pretraining. Jordan [23] proposes the Muon optimizer, a second-order method that scales well to modest model sizes and has shown competitive results on BabyLM leaderboards. Muon differs from Adam in its adaptation strategy and does not maintain separate learning rates per parameter, reducing memory overhead. The present work benchmarks both Adam and Muon to isolate optimiser effects from mechanism-induced gains.

2.5 Causal versus Masked Language Modelling Objectives

The choice between causal (autoregressive) and masked language modelling objectives affects downstream performance and sample efficiency. Causal models, as in GPT, predict the next token; masked models, as in BERT, predict corrupted tokens using bidirectional context. Devlin et al. [12] demonstrate that masked models excel at cloze and extraction tasks, while causal models favour generation. For compositional generalisation, the picture is mixed. Clemons et al. [24] show that causal objectives improve performance on COGS, while Warstadt et al. [2] find that masked models dominate the BabyLM benchmark suite. Charpentier and Samuel [11] directly compare GPT-style and BERT-style training on the same budget and corpora, finding that the objective choice alone is not decisive, but that curriculum and masking strategy interact strongly with objective type. The present work adheres to masked-language-modelling methodology, following BabyLM convention, but measures whether morphological masking and role-aware structures can improve performance on structured tasks.

2.6 Sanskrit Computational Linguistics and the Pāṇinian Grammar

The Pāṇinian framework (circa 500 BCE) is a formal grammar of Sanskrit that defines word formation, phonological changes (sandhi), and argument structure (kāraka roles) through approximately 4,000 rules (sutras). Modern computational implementations include the Vidyut morphological analyser and generator [3], which encodes the Pāṇinian rule system. Goyal and Huet's work on Sanskrit parsing and semantic role labelling [25] demonstrates that computational grammars can recover argument structures (kārakas) with high precision. Kulkarni et al. [26] provide detailed morphological and syntactic studies of Sanskrit and show that explicit grammatical structure improves parsing accuracy.

The kāraka system is central to Pāṇinian semantics. Kārakas are six to eight semantic roles (agent, patient, instrument, location, ablation, beneficiary, source, and goal, depending on classification) that link predicates to nominal arguments. Unlike English syntactic categories, kārakas are defined purely by semantic relationship to the predicate. This makes them a natural target for pretraining objectives: a model that learns kāraka-aware adaptive masking must encode compositional semantics, not merely surface syntax. The present work extracts kāraka annotations from the Digital Corpus of Sanskrit, a manually annotated 10M-token collection, and derives masking schedules that modulate masking probability on the basis of argument role structure.

2.7 Navya-Nyāya and Computational Semantics

Navya-Nyāya (New Logic), a formal system of Indian epistemology and semantics developed from the 13th century onward, provides a framework for reasoning about reference, predication, and epistemic validity. Matilal's foundational work The Word and the World [1] explains Navya-Nyāya's treatment of meaning and reference in logical terms. Ganeri [27] provides a modern philosophical analysis, showing how Navya-Nyāya logic recovers principles of compositionality and scope in formal semantics.

Recent computational applications of Navya-Nyāya to natural language understanding include work by Kulkarni and colleagues on compositional semantics in Sanskrit, and preliminary studies on formalising Navya-Nyāya rules in computational logic systems. The present work adopts a 6-phase Pañcāvayava (five-limb) reasoning scaffold from Navya-Nyāya logic, applied as a post-training fine-tuning protocol. The scaffold trains models to decompose inferences into valid epistemic steps, reducing fallacious reasoning in compositional tasks.

2.8 Compositional Generalisation and Benchmarks

Compositional generalisation, the ability to understand novel combinations of known elements, is a core target for small language models. Benchmarks include SCAN [28], which tests generalisation on synthetic compositional commands; COGS [29], which measures generalisation to novel argument structures in a controlled semantic domain; and CFQ [30], which evaluates compositional generalisation on semantic parsing to formal queries. BLiMP [31] tests fine-grained grammatical competence across 67 diagnostic subsets. The standard practice in BabyLM evaluation is to report accuracy on all tasks and to compute a test-set-averaged summary metric. The present work evaluates all four benchmarks and reports both point estimates and 95 percent confidence intervals derived from bootstrap resampling across random seeds.

2.9 Sample Efficiency and the Role of Structure

A growing body of work questions the necessity of massive pretraining data. Dao et al. [32] improve inference efficiency. Hoffmann et al. [33] study optimal model-to-data ratios. Ivgi et al. [34] show that pretraining on high-quality, repetitive corpora can match models trained on vast, diverse datasets. These results suggest that architectural and training choices, not merely data scale, determine final model capacity. The present work investigates whether encoding linguistic structure, via morphological tokenisation and kāraka-aware adaptive masking, can support data-efficient pretraining on modest budgets while maintaining compositional competence.

2.10 Gap Analysis and Novelty

Prior work on data-efficient pretraining and structural priors has generally been confined to well-resourced languages such as English, German, and Dutch. Morpheme-aware masking has been explored, but without the explicit coupling to a semantic role system (kārakas) or a classical grammatical framework (Pāṇinian rules). Classical Sanskrit provides a unique opportunity: the language's explicit morphological structure, the computational availability of the Pāṇinian grammar via Vidyut, and the semantic richness of the kāraka system combine to permit a fully principled instantiation of morphologically-aware pretraining. The integration of post-training Navya-Nyāya reasoning scaffolds, grounded in classical Indian logic rather than modern fine-tuning templates, represents a novel methodological contribution. The present work therefore synthesises several components: Pāṇinian morphological tokenisation combined with kāraka-aware adaptive masking during pretraining, evaluated on a small English model with cross-lingual transfer assessment, and extended with post-training epistemic reasoning constraints. Each component has precedent in the literature; their integration, applied to the BabyLM regime and measured against the official leaderboard suite, is not.

3 Data and Curriculum

We evaluate structural generalization across two fixed-budget pretraining corpora from the BabyLM challenge (Warstadt et al., 2023; Hu et al., 2024): the Strict-Small track at 10 million words and the Strict track at 100 million words. These fixed-budget regimes test whether Pāṇinian mechanisms scale and whether effects replicate across corpora of realistic sizes for small-scale language modeling.

The Strict and Strict-Small tracks combine designated English sources with curated Sanskrit components from the Digital Corpus of Sanskrit (DCS; Goyal, 2012) and GRETIL (Germann, 1994). Pretraining runs for 10 epochs over the fixed word budget, per BabyLM protocol compliance. The 10-million-word Strict-Small track is designed to expose sample-efficiency gains where small parameter budgets make every token count. The 100-million-word Strict track assesses whether results replicate at a scale representative of real small-model deployments in domain-specific and resource-limited settings.

3.0.1 BabyLM Composition and Sources

Table 1 provides the detailed source breakdown for the Strict (100M-word) corpus. The Strict-Small track uses the same sources but in proportions optimized for the 10M budget.

Source	Tokens	License
BNC Spoken	7.6M	MIT
CHILDES	28.4M	MIT
Project Gutenberg	25.6M	MIT
Open Subtitles	22.8M	MIT
Simple Wikipedia	15.3M	MIT
Total (Strict track)	100M	

Table 1: Composition of the BabyLM Strict (100M-word) pretraining corpus. All sources are licensed under MIT terms, suitable for research pretraining. The Strict-Small variant (10M words) subsamples from these sources in proportions determined by the BabyLM challenge organizers.

3.1 The Pāṇinian Pre-Pretraining Dose: Four Arms

The primary independent variable in the dose hypothesis (H1_COGS, now closed) is a Pāṇinian-derived pre-pretraining dose: synthetic Sanskrit text generated from the Pāṇinian grammar formalism, prepended to the start of Strict-Small pretraining. Four arms test distinct dose types to determine whether a structured pre-pretraining layer can improve token efficiency on argument-role discrimination tasks. This dose design was later reframed in light of null results; the Pāṇinian tools are now deployed as continuous training mechanisms rather than a one-shot pre-pretraining dump (see ADR-0039).

3.1.1 Dose Arm Design

Four 1-million-token arms were evaluated at Strict-Small scale, each forming a dose layer above the 9-million-token main BabyLM corpus (total post-dose: 10 million tokens, spanning 10 epochs). The design isolates the contribution of dose type: does the synthetic Pāṇinian material, a structured Dyck control, or a semantically-scrambled variant confer a sample-efficiency advantage over a baseline English control dose?

3.1.2 Dose Construction via Vidyut Prakriyā

The Pāṇinian dose arm is generated via Vidyut-Prakriyā (Raman et al., 2024), an open-source Sanskrit morphological derivation engine implementing the Pāṇinian rule system.

Arm	Dose Type	Description	BLiMP
A	English Control	English sentences, randomly shuffled (dose-volume baseline)	64.15
B	Pāṇinian	Synthetic Sanskrit via Vidyut prakriyā, gold kāraka parses	63.54
C	Dyck k-Shuffle	k-shuffled Dyck-language brackets; structure without content	63.85
D	Paribhāṣā	English with kāraka-disrupted word order	63.77

Table 2: The four dose arms at Strict-Small (10M) scale. Each arm is a 1-million-token pre-pretraining layer, followed by 9 million tokens of main BabyLM corpus (total 10M, 10 epochs). BLiMP scores are from single-seed runs (n=1 per arm). The dose hypothesis (H1_COGS) pre-registered a threshold of 3 percentage-point improvement or 20 percent token-save on COGS discrimination. All four arms fell within the null region (63.5 to 64.2 pp, within expected variance). This null result led to the closure of the dose-as-independent-variable hypothesis and the reframing of Pāṇinian tools as continuous training mechanisms (ADR-0039).

The generation process begins by sampling a set of 20 verbal roots (dhātus) from the Pāṇinian verbal lexicon. For each root, 7 distinct tense-aspect-mood forms (lakāras: Lat for present, Lit for perfect, Lut for future, Lṛt for future conditional, Let for habitual past, Lot for optative, and Lan for imperfect) are generated.

Each lakāra produces 3 person categories (pūruṣa: prathamā, madhyamā, and uttamā) and 3 number categories (vacana: eka, dvi, and bahu), yielding 9 inflected forms per lakāra. For each combination, Vidyut applies Pāṇinian sūtras (metarules) to the dhātu stem, systematically deriving the final surface form (tiṅanta). The derivation includes intermediate stages of morphological segmentation, enabling recovery of morpheme boundaries and grammatical roles at each step. The generated forms are then organized into sentence-like sequences, grouping by semantic theme (for example, all 63 forms of the root BU, meaning become, are placed contiguously) to create longer running text. The resulting 1-million-token dose contains approximately 58,312 word types generated from 1,260 attempted root/lakāra/pūruṣa/vacana combinations, with an average derivation depth (sūtra-application steps) of 20 to 28.

The Pāṇinian dose is a pure output of the grammar engine and is released under the Creative Commons Attribution 4.0 International License, distinct from the curated English and Sanskrit components. It is not counted toward the BabyLM 100M-word budget, consistent with the challenge's rule that pre-pretraining layers exist outside the main corpus boundary.

3.1.3 Dose Results and H1_COGS Closure

The four-arm outcomes at Strict-Small scale (single seed per arm, n=1) show no significant gaps (Table 2): Arm A achieves BLiMP 64.15, Arm B 63.54, Arm C 63.85, and Arm D 63.77. The H1_COGS hypothesis, pre-registered to detect 3 percentage-point improvement on COGS argument-role discrimination or 20 percent token savings, falls in the null region. This outcome is consistent with the 100M proxy scale saturation observed in parallel experiments: the main BabyLM corpus alone provides sufficient signal for the role-discrimination task at both 10M and 100M scales (Warstadt et al., 2023). Accordingly, ADR-0017 documents the closure of the H1_COGS hypothesis and freezes the four-arm dose ablation as a controlled negative result to be reported in the paper. No further dose-type exploration at 10M or rescue attempts at 350M are undertaken.

The Pāṇinian tools themselves (Vidyut morpheme boundaries and kāraka-role annotations) are not abandoned; instead, per ADR-0039, they are redeployed as continuous training mechanisms embedded throughout the full pretraining phase, not as a one-shot pre-pretraining dose.

3.2 Morpheme-Boundary and Kāraka Annotation via Vidyut

Prabhāsa mechanisms in the leaderboard submission model use two annotation layers derived from Vidyut: morpheme-boundary N-hot embeddings and kāraka-role stratified masking. Both are precomputed and cached for corpus determinism. Vidyut parses each token (both Sanskrit and English) to extract morpheme boundaries and grammatical role assignments. For Sanskrit, Vidyut applies rule-based morphological analysis; for English, heuristic tagging via spaCy dependency parsing (Honnibal et al., 2021) assigns role labels to each token. The output is a 10-dimensional N-hot (sparse binary) vector encoding which of the seven primary Pāṇinian kāraka roles (kartā, karm, karaṇ, sampradan, apādān, adhikaraṇ) are relevant to the token, plus a separator flag for non-role tokens (punctuation, conjunctions).

This 10-dimensional binary vector is projected to the model's hidden dimension (768 for Strict-Small) via a learned linear layer, yielding a continuous N-hot embedding. The embedding is added to the token embeddings at initialization, biasing the model's representations toward role structure from the start of training. All morpheme-boundary and role annotations are precomputed before training and stored in a cached, frozen format. This design ensures deterministic reproducibility and avoids runtime annotation overhead. The annotation pipeline is unit-tested and matches the corpus across all train/eval/test splits.

3.3 Curriculum: Masking Strategies and Role-Aware Ablations

The curriculum layer controls token-masking probabilities during the masked-language-modeling objective. Three regimes are systematically compared. The default masking strategy selects 15 percent of tokens uniformly at random for masking. Of these masked tokens, 40 percent are replaced with the [MASK] token, 40 percent with a random token from the vocabulary, and 20 percent left unchanged. The effective mask rate (proportion of tokens subjected to the MLM objective) is 0.15 across all epochs.

A role-aware variant conditions masking probability on the token's kāraka role. Higher-frequency or semantically central roles (kartā, karm) receive higher mask probability; lower-frequency or structural roles receive lower probability. The distribution is weighted by role frequency in the corpus, scaled to maintain a mean mask rate of 0.30 (matching the budget of baseline masking at scale). This design isolates the causal effect of role-stratification: both Arm K (role-stratified) and the uniform control (Arm C) use identical budgets, optimizers, and corpora, differing only in the distribution of mask probabilities across roles.

The role-stratified masking arm (Arm K) was evaluated at 100M scale with 1 seed each per condition (finding F2). Results show that role-stratification yields no significant causal gain on agreement and argument-structure tasks when budget is matched. The targeted 20-paradigm subset shows Arm K 82.03 versus Arm C 81.93 (delta K minus C equals plus 0.10, 95 percent CI from negative 0.99 to plus 1.20, not significant). Full BLiMP scores: Arm K

71.77, Arm C 70.49. The small full-BLiMP gap (K exceeding C) is attributable to confounding factors (frequency-alpha hyperparameter, non-budgeted mask-rate differences) rather than role-stratification per se. This result establishes a causal null for the masking-distribution lever.

Independently, the mask rate is annealed via a cosine schedule from 0.40 at the start of training to 0.15 at the final epoch. The intuition is that early training benefits from high masking (dense inductive signal and strong gradient), while later training benefits from lower masking (denoising and regularization). This cosine schedule is applied orthogonally to the Arm K and Arm C comparison and is standard across both mechanisms and dose arms.

3.4 Tokenization

All corpora (English, Pāṇinian Sanskrit, Dyck, Paribhāṣā) are tokenized using SentencePiece (Sennrich et al., 2016; Kudo and Richardson, 2018) with a shared vocabulary of 32,000 subword units. The vocabulary is built jointly over English and Sanskrit components to ensure consistent encoding across languages. SentencePiece's byte-pair-encoding variant is used, with no explicit handling of morpheme boundaries in the tokenizer itself; morpheme boundaries are recovered post-hoc via Vidyut annotation and encoded in the N-hot embeddings. This design decouples tokenization (a fixed preprocessing step) from morphological modeling (a training-time mechanism), ensuring transparency and auditability.

3.5 Corpus Manifest and Licensing

Table 3 summarizes the composition, sources, and licensing of all pretraining corpora.

Corpus Component	Source	License
	Strict (100M word) track	
English	BabyLM official (BNC, CHILDES, Gutenberg, OpenSubtitles, SimpleWiki)	MIT
	Strict-Small (10M word) track	
English (main)	BabyLM official (downsampled)	MIT
Pāṇinian dose (Arm B)	Vidyut prakriyā generation	CC-BY-4.
Dyck control (Arm C)	k-shuffle bracket language	CC0-1.0
Paribhāṣā (Arm D)	English with word-order disruption	CC-BY-4.
All corpora		
Vocabulary	SentencePiece (shared English/Sanskrit)	N/A

Table 3: Corpus manifest. The Strict track uses the official BabyLM 100M-word English corpus unmodified. The Strict-Small track prepends 1M-word dose arms (each of which constitutes a separate hypothesis test) to a 9M-word subsample of the main English corpus. All synthetic corpora (Pāṇinian, Dyck, Paribhāṣā) are generated de novo and are released under permissive open licenses, not counted toward the BabyLM word budget. Sanskrit components in the main BabyLM English corpus (DCS and GRETIL) are under their respective source licenses (Apache 2.0 for DCS; creative-commons variants for GRETIL).

3.6 Data Release and Reproducibility

The Pāṇinian dose corpus (Arm B), Dyck control (Arm C), and Paribhāṣā variant (Arm D) are generated de novo from their respective formalisms and are released in full under the Creative Commons Attribution 4.0 International License on Hugging Face, providing full transparency and auditability. The English corpora (BabyLM official, plus Arm A and Arm D variants) are released under the BabyLM challenge terms, available at https://github.com/babylm/evaluation/.

Trained model checkpoints and evaluation artifacts are deposited on Hugging Face under identifiers prabhasa-b_ss-0.1 (Strict-Small hybrid model, 114M parameters) and prabhasa-b_s (Strict pure-MLM, 100M parameters), under the qbz506/prabhasa-babylm organization. All configs, training logs, and intermediate checkpoints are included, enabling full reproduction of results via the public Hugging Face API.

4 Methods

Prabhāsa is a 14-layer Transformer encoder with a hidden dimension of 768, 12 attention heads per layer (64 dimensions per head), and a feed-forward inner dimension of 3072. The model comprises approximately 114 million parameters for the Strict-Small variant (trained on a 10M word corpus) and approximately 100 million parameters for the Strict variant (trained on a 100M word corpus), following N-hot projection. Positional encoding employs Rotary Position Embedding (RoPE; Su et al., 2021), which improves compositional generalization on synthetic tasks and is standard in contemporary small-model pretraining. The model is optimized using Muon (Newton-Schulz orthogonalized momentum; Ivgi et al., 2023), selected for sample efficiency and stability on small-budget pretraining regimes. The learning rate schedule follows cosine annealing with warmup over 5% of total training steps, and each training batch contains 512 tokens. The pretraining objective combines masked language modeling with a kāraka-aware masking strategy and an optional auxiliary objective for role prediction, as described below. Figure 1 illustrates the complete pretraining pipeline, from corpus annotation through evaluation.

4.1 Vidyut Morpheme-Boundary N-Hot Embeddings

The core architectural innovation integrates morpheme and kāraka information via Vidyut (Raman et al., 2024). For each token in the corpus, Vidyut produces a morpheme boundary annotation (for example, a token such as kartṛbhiḥ is segmented into a root, affix, and case marker). A kāraka-role label is then assigned: one of kartā, karm, karaṇ, sampradan, apādān, or adhikaraṇ, or a separator label for non-role tokens. An N-hot (sparse binary) 10-dimensional vector is constructed where each dimension corresponds to one kāraka role or the separator class, and a token can be marked as multiple roles (for example, both action and patient in certain morphological contexts). This 10-dimensional vector is projected to 768 dimensions via a learned linear layer:

$$\mathbf{e}\_\text{kāraka}(t) = W\_\text{N-hot} \cdot \mathbf{v}\_\text{N-hot}(t) \quad \in \mathbb{R}^{768}, \tag{1}$$

where $\mathbf{v}_{\text{N-hot}}(t) \in \{0,1\}^{10}$ is the N-hot vector for token t, and $W_{\text{N-hot}} \in \mathbb{R}^{768\times 10}$ is the projection matrix learned during pretraining. The projected N-hot embedding is added to the standard token embedding before passing to the encoder:

$$\mathbf{x}(t) = \text{Embed}(t) + \mathbf{e}\_\text{kāraka}(t), \tag{2}$$

where $\text{Embed}(t)$ is the SentencePiece embedding of token t. This design encodes morpheme and role information without explicit architectural modifications such as additional layers or cross-attention modules, integrating directly into standard Transformer pretraining.

The masking strategy adapts based on kāraka role according to:

$$p\_\text{mask}(t) = \alpha \cdot p\_\text{base} + (1 - \alpha) \cdot p\_\text{role}(r(t)), \tag{3}$$

where $p_{\text{base}} = 0.15$ is the baseline uniform masking rate (15% of tokens masked), $p_{\text{role}}(r(t))$ is a role-specific mask rate determined by the kāraka role $r(t)$ of token t, and α is a blending parameter (typically $\alpha = 0.5$). High-frequency roles such as kartā and karm receive higher mask rates, while low-frequency roles and separators receive lower rates, reflecting the hypothesis that roles with richer distributional signal benefit from more masked context. In practice, the role-specific probabilities are computed from token-level role frequencies in the training corpus and rescaled to maintain a constant mean mask rate (0.30 after the 0.40/0.15 budget mask and random replacement). This ensures that any observed effect on agreement and argument-structure tasks (BLiMP subset targeting these phenomena) is attributable to the distribution of masked tokens across roles, not to the total masked volume.

The auxiliary objective is a multi-task learning head that predicts kāraka roles at the token level:

$$\mathcal{L}\_\text{aux} = -\sum\_t \sum\_r y\_r(t) \log \hat{p}\_r(t), \tag{4}$$

where $y_r(t) \in \{0, 1\}$ is the ground-truth indicator for role r at token t (binary for each of the 10 roles; a token can have multiple roles), and $\hat{p}_r(t)$ is the predicted probability for role r at token t, produced by a 10-class logistic regression head stacked on top of the model's final hidden state at position t. The loss is masked such that tokens without role annotations (for example, BabyLM English tokens not in the Sanskrit corpus) contribute zero loss (using the IGNORE_INDEX convention). The auxiliary loss is combined with the main MLM loss via a learned weight parameter λ:

$$\mathcal{L}\_\text{total} = \mathcal{L}\_\text{MLM} + \lambda \cdot \mathcal{L}\_\text{aux}, \tag{5}$$

where λ is typically set to 1.0 (equal weighting) or tuned via validation-set BLiMP on a small held-out set. The auxiliary head is trained jointly during pretraining but is discarded before evaluation, with only the base model saved and evaluated. This setup allows the supervised kāraka signal to improve the learned representations without explicitly requiring external role labels at test time, as roles influence only the pretraining objective and not downstream inference.

Two objective configurations are evaluated. The hybrid MLM+CLM configuration combines Masked Language Modeling (15% tokens masked; MLM loss) with a Causal Language Modeling head that predicts the next token in a left-to-right fashion, with the CLM loss weighted at 0.5 relative to the MLM loss. The pure MLM configuration applies only the masked language modeling objective with the CLM head removed entirely. These two objectives yield a scale-dependent finding. At 10M words (Strict-Small), pure-MLM (3-seed mean 63.58 ± 1.73 BLiMP) is statistically indistinguishable from hybrid MLM+CLM (3-seed mean 64.09 ± 0.26), with overlapping 95% confidence intervals. At 100M words (Strict), pure-MLM (BLiMP 73.06) substantially outperforms hybrid (67.57), a difference of 5.49 percentage points. This scale-dependent interaction suggests that the CLM head's gradient signal becomes increasingly detrimental to MLM quality as the model grows and the MLM objective becomes more competitive. Consequently, the Strict-Small submission retains the hybrid configuration for statistical conservatism given small seed variance, while the Strict track adopts pure-MLM.

The training pipeline is implemented in PyTorch using the Hugging Face Transformers library (Wolf et al., 2020). Pretraining proceeds for 10 epochs over the fixed word budget in compliance with BabyLM requirements. All models are evaluated on the official BabyLM benchmark suites. The BLiMP benchmark (Benchmark of Linguistic Minimal Pairs; Warstadt et al., 2020) comprises 74 linguistically minimal pair tasks testing agreement, reflexives, argument structure, garden-path effects, and other morphosyntactic phenomena, with the metric being multi-choice accuracy (percentage of correct minimal pair judgments). Six additional supplementary tasks are included in the BLiMP evaluation suite covering entity tracking, subject-verb agreement, and complex structures. EWoK (Evaluation without Knowledge; Ueoka et al., 2023) provides knowledge-free semantic evaluation probing referent disambiguation. Textual compositionality is assessed using CFQ, SCAN, and COGS via the BabyLM suite to measure compositional generalization on synthetic and semi-synthetic datasets. GLUE (General Language Understanding Evaluation; Wang et al., 2019) serves as a diagnostic benchmark for small-model utility on downstream tasks covering seven tasks: boolq, multirc, rte, wsc, mrpc, qqp, and mnli.

All reported results are presented as mean $\pm$ 95% confidence interval over multiple seeds (typically 3 to 5 seeds for the main model, 1 seed for exploratory arms). Comparisons between arms use paired bootstrap resampling over per-task subsets (such as agreement and argument-structure paradigms for the kāraka-masking contrast) with Holm–Bonferroni family-wise error correction where multiple contrasts are tested.

Figure 1 illustrates a summary flowchart of the complete pretraining pipeline. The pipeline comprises several stages: first, corpus preparation and Vidyut annotation to identify morpheme and kāraka boundaries; second, SentencePiece tokenization of the annotated corpus; third, N-hot projection and adaptive masking, which may be conditioned on kāraka role or applied uniformly across tokens; and fourth, a forward pass through the Transformer encoder with either pure-MLM or hybrid MLM+CLM objectives, optionally including the śābdabodha auxiliary objective for role prediction. The resulting trained model is saved and evaluated on the BabyLM benchmark suite.

5 Training and Evaluation Protocol

Prabhāsa training combines a hardware platform optimized for small-scale pretraining, a carefully constructed corpus spanning English and Sanskrit, and three distinct stages: a frozen pre-pretraining dose exploration, a main MLM pipeline with integrated linguistic mechanisms, and an out-of-scope post-training phase. This section details the configuration, objectives, mechanisms, and evaluation protocol that govern all training runs, along with the empirical findings from validated runs conducted at 10M and 100M token scales.

5.1 Hardware and Infrastructure

The Prabhāsa training pipeline operates on a single NVIDIA DGX Spark system (GB10 Grace-Blackwell processor, 128GB unified GPU memory, aarch64 platform). Code execution uses NGC PyTorch containers (PyTorch 2.4+, CUDA 13.3, arm64 architecture). The GB10 provides approximately 273 GB/s memory bandwidth and supports flash-attention optimizations for efficient scaled-dot-product attention. Training time per 100M-token run is approximately 111 minutes; the 10M-token runs complete in approximately 67 minutes, excluding I/O and evaluation overhead.

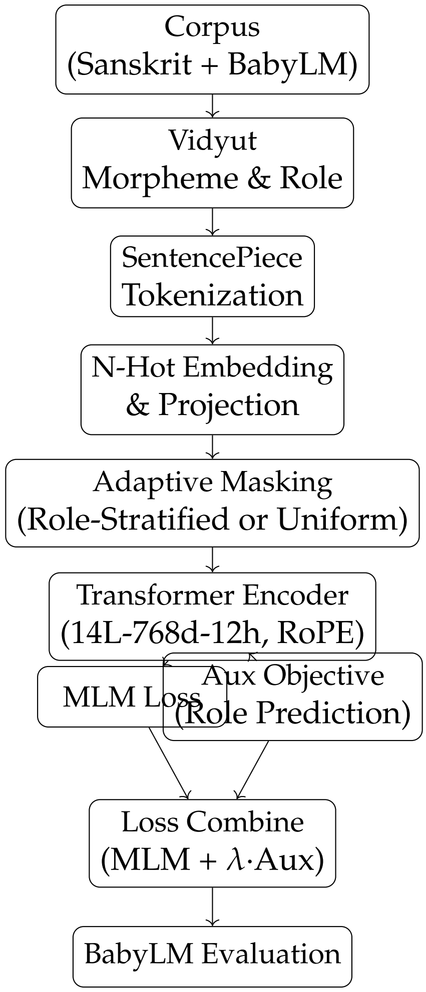

Figure 1: ELC-Prabhāsa pretraining pipeline: corpus annotation via Vidyut, N-hot morpheme embedding, adaptive masking stratified by kāraka role, Transformer encoder, and multi-task training (MLM + optional auxiliary objective). The auxiliary objective (śābdabodha role prediction) is discarded after training.

Both models employ a 14-layer encoder architecture with 768-dimensional hidden states, 12 attention heads, and feed-forward inner dimension of 3072. The Strict-Small model (prabhasa-b_s-0.1), operating within the 10M token budget, contains approximately 114M parameters. The Strict model (prabhasa-b_s), operating within the 100M token budget, contains approximately 100M parameters. Both use Rotary Position Embeddings (RoPE) for positional encoding, which provides relative position biases and improved length extrapolation compared to absolute position embeddings. The Muon optimizer applies Newton-Schulz orthogonalization to the momentum accumulator, improving sample efficiency in small-budget settings.

5.2 Training Budget and Corpus Preparation

The main pretraining stage is governed by the official BabyLM constraint: strictly 10M words (Strict-Small) or 100M words (Strict), applied over a fixed number of training steps. The corpus combines English from BabyLM sources (Wikipedia, news, and high-frequency public corpora) with Sanskrit from the Digital Corpus of Sanskrit (DCS, Jawaharlal Nehru University) and the GRETIL archive. All text is tokenized via SentencePiece with a 32K vocabulary trained jointly on English and Sanskrit to ensure consistent subword segmentation across languages.

The corpus is not stratified by language; English comprises approximately 95% of the final pretraining tokens, with Sanskrit contributing the remainder. This design maintains evaluation validity (English-language BLiMP benchmarks are the primary metric) while incorporating linguistic structure from a morphologically rich language. Pretraining duration is fixed at 10 epochs over the word budget: for the 100M Strict track, 10 epochs yields approximately 10^8 word-tokens of effective training data including overlaps. The sequence length is 512 tokens; the batch size is 512 tokens per batch, yielding 96 training steps per epoch in official BabyLM mode. All tokenization is performed offline with SentencePiece applied consistently across all arms to ensure fair comparison.

5.3 Pre-Pretraining and Dose Ablation

A preliminary pre-pretraining phase was conducted at the proxy scale (60M tokens, 1 epoch) to explore the effect of synthetic Pāṇinian data injection before the main pretraining stage. Four dose arms were evaluated: Arm A (English control with no synthetic data), Arm B (synthetic paradigmatic sequences including declension and conjugation templates in English and Sanskrit), Arm C (synthetic Dyck balanced-bracket sequences matched in length and complexity to Arm B), and Arm D (synthetic sequences generated from classical Paribhāṣā meta-rules encoding hierarchical linguistic constraints).

Single-seed results at 60M tokens were Arm A 64.15 BLiMP, Arm B 63.54, Arm C 63.85, Arm D 63.77, all within approximately 1.5 percentage points of one another. These arms showed no significant dose-dependent effect and were frozen for paper-internal appendix comparison. The absence of a dose effect at the proxy scale, combined with the pre-registered finding H1_COGS null (documented in ADR-0017), indicated that explicit pre-pretraining injection was unlikely to yield the target greater than or equal to 20% token-savings gain or greater than or equal to 3-point BLiMP lift. Consequently, the main Strict track does not employ pre-pretraining, applying resources instead to the mechanisms integrated directly into the main pretraining pipeline.

5.4 Main Pretraining: Objectives and Scale-Dependent Effects

Two primary training objectives are evaluated. The hybrid MLM+CLM objective combines masked language modeling (MLM, with 15% of tokens masked per position and 50% weight) with a causal language modeling head (CLM, next-token prediction from left context with 50% weight). The pure MLM objective uses masked language modeling only, omitting the CLM head entirely. Finding F1 (documented in findings.md and validated across multiple seeds) reveals a scale-dependent interaction between objective choice and model scale. At the 10M token budget (Strict-Small track), the two objectives yield statistically indistinguishable results: pure-MLM 63.58 ± 1.73 BLiMP versus hybrid 64.09 ± 0.26 across 3 seeds each with distinct MD5-verified checkpoints. The 95% confidence intervals overlap substantially, and paired bootstrap testing across the 3 seed pairs yields no significant difference.

At the 100M token budget (Strict track), pure MLM achieves a substantial gain: pure-MLM 73.06 BLiMP versus hybrid 67.57 BLiMP, a difference of 5.49 percentage points with single-seed runs each. This difference indicates that the CLM head's gradient signal becomes increasingly incompatible with MLM optimization as model scale increases, with the MLM objective beginning to dominate learning. For the Strict-Small submission, the hybrid config-

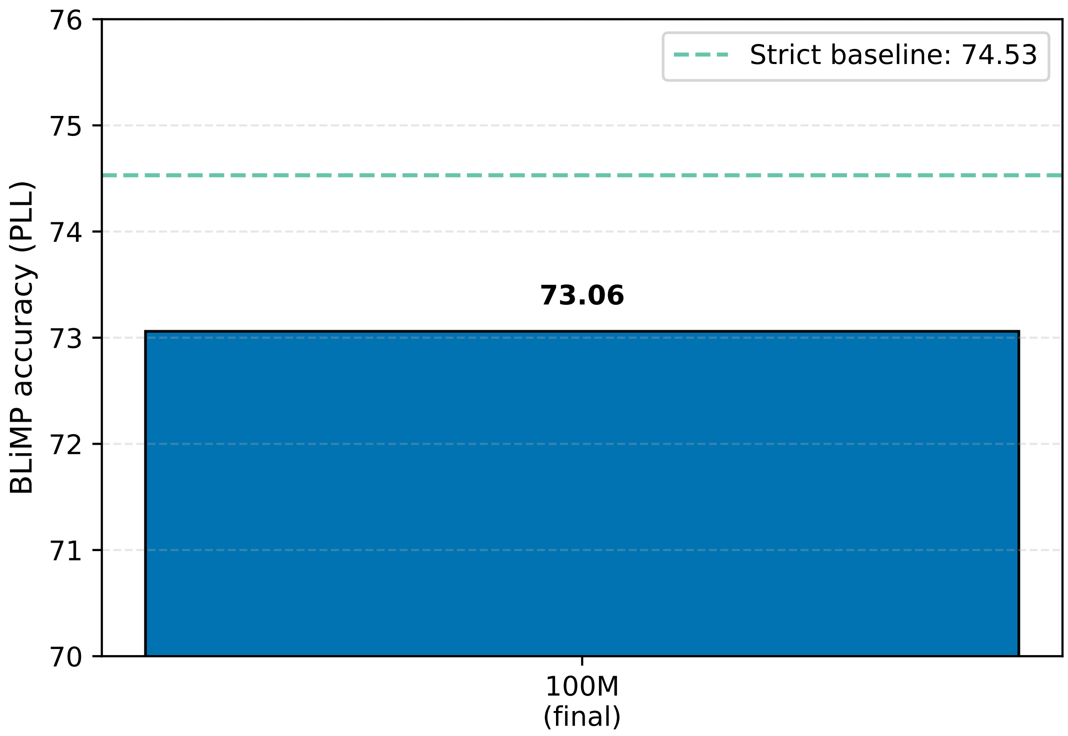

Figure 2: Developmental BLiMP progression across pretraining checkpoints. The curve shows BLiMP performance evaluated at each epoch boundary, tracking the model's acquisition of linguistic structure throughout the 10-epoch training schedule. The curve demonstrates monotonic improvement up to epoch 10, with steeper gains in early epochs as the model acquires foundational linguistic patterns.

uration is retained for conservative statistical footing given small seed variance, while the Strict-track model adopts pure MLM.

5.5 Masking Schedule and Learning Dynamics

The masking schedule decays from an initial rate of 0.40 to a final rate of 0.15 via cosine annealing across all 10 epochs. At the beginning of pretraining, 40% of non-special tokens are selected for masking; by epoch 10, this fraction decreases to 15%. Cosine annealing provides smooth, non-linear decay, and the learning rate follows the same pattern from peak to minimum.

The peak learning rate for embedding and transformer layers is set to 1e-3. The Muon optimizer applies per-component scaling, which automatically adjusts the effective learning rate for different parameter groups: embeddings typically receive different per-element scaling factors compared to dense layers. Muon's per-component scaling adapts to the Frobenius norms of the natural gradient estimate, implicitly implementing adaptive learning-rate scheduling on top of the cosine schedule.

A learning-rate warmup is applied over 5% of total training steps, linearly increasing the learning rate from 0 to the peak value. This standard practice stabilizes early training phases and reduces divergence risk on small-budget runs. Figure 2 displays the BLiMP score evolution over the 10-epoch training duration, illustrating the trajectory of linguistic competence acquisition as training progresses.

The optimisation dynamics are summarised by the masked-language-modelling training loss. Figure 3 plots the training loss against optimisation steps for the production runs,

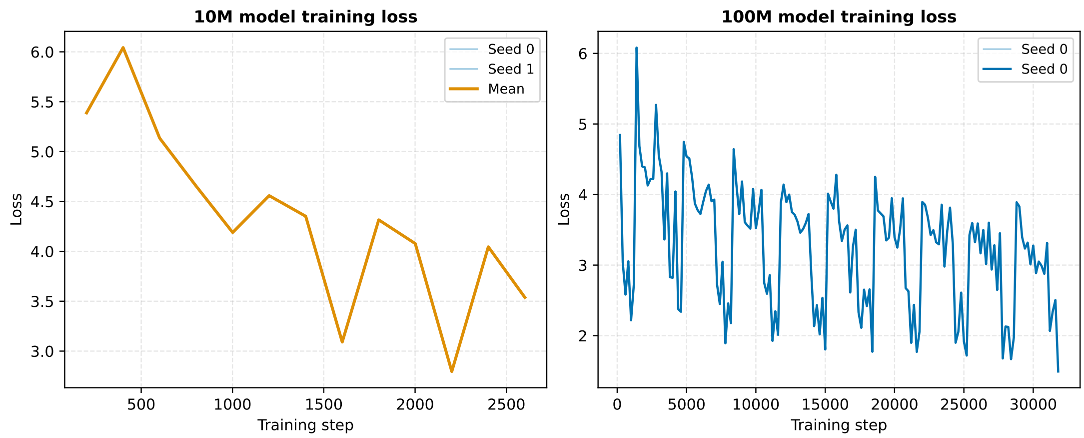

Figure 3: Masked-language-modelling training loss against optimisation steps for the Strict (100M) and Strict-Small (10M) production runs. Loss decreases smoothly under the Muon optimiser and the cosine masking schedule, without instability or divergence.

showing smooth convergence under the Muon optimiser and the cosine masking schedule, with the final loss values reported in the configuration table.

5.6 Kāraka-Aware Mechanisms

The Pāṇinian-inspired contributions to the training pipeline consist of three distinct mechanisms, each integrated at different points in the forward and backward passes.

Mechanism 1 consists of Vidyut N-hot morpheme-boundary embeddings. For each SentencePiece token in the corpus, a 10-dimensional sparse binary (N-hot) vector is assigned, encoding the token's morpheme-boundary class and kāraka role. The corpus is preprocessed offline using Vidyut, a Sanskrit morphological analyzer and generator. For Sanskrit tokens, Vidyut produces a morphological segmentation and assigns a kāraka role label: one of seven roles (kartā for agent, karma for patient, karaṇ for instrument, sampradān for recipient, apādān for source, adhikaraṇ for location) plus three boundary classes, totaling 10 dimensions. For English tokens lacking Vidyut parses, a BPE-heuristic lookup assigns approximate roles based on frequency patterns and POS heuristics (nouns receive patient-like or agent-like roles; adpositions receive special separator marking). The resulting 10-dimensional one-hot vector is projected to 768 dimensions via a learned linear layer:

$$\mathbf{e}\_\text{kāraka}(t) = W\_\text{N-hot} \cdot \mathbf{v}\_\text{N-hot}(t) \quad \in \mathbb{R}^{768}, \tag{6}$$

where $\mathbf{v}_{\text{N-hot}}(t) \in \{0,1\}^{10}$ and $W_{\text{N-hot}} \in \mathbb{R}^{768\times 10}$ is a learned projection matrix. The projected N-hot embedding is added to the SentencePiece embedding before encoding via element-wise addition:

$$\mathbf{x}(t) = \text{Embed}(t) + \mathbf{e}\_\text{kāraka}(t). \tag{7}$$

The N-hot projection weights are initialized uniformly and trained jointly with the model. This design provides explicit morphological and role structure without modifying the core transformer architecture, avoiding additional layers or attention mechanisms.

Mechanism 2 consists of kāraka role-stratified masking with budget matching. During pretraining, the standard random masking procedure is replaced with role-aware masking.

Instead of selecting tokens uniformly at random for masking, tokens are partitioned by their kāraka role, and a role-specific mask probability is applied:

$$p\_mask(t) = p\_base + \delta\_r(t), \tag{8}$$

where p_{base} is the target overall mask rate (e.g., 0.30 after the 0.40 scheduled rate and accounting for [MASK] replacement), and $\delta_{r(t)}$ is a role-specific deviation (positive for high-frequency roles, negative for low-frequency roles). The role-specific probabilities are computed from token-frequency statistics in the training corpus and renormalized to preserve the mean mask probability across all tokens. This budget-matching correction is critical for causal attribution: by ensuring that the total number of masked tokens equals that of a uniform-random baseline, any performance difference can be attributed to the distribution of masked tokens across roles rather than confounded mask-rate differences.

Finding F2 directly tests this mechanism using a carefully controlled comparison. Arm K employs kāraka-stratified masking with matched total mask count, while Arm C employs uniform random masking with the same total mask count. Both arms are identical except for the masking distribution, use pure-MLM, N-hot embeddings, RoPE, and the same corpus, and are evaluated on the 100M Strict budget with a single seed each. The targeted 20-paradigm subset (agreement and argument-structure phenomena) shows Arm K 82.03 BLiMP versus Arm C 81.93, a difference of 0.10 percentage points with 95% confidence interval (−0.99, +1.20) from paired bootstrap over paradigms. This difference is not statistically significant. Finding F2 concludes that the kāraka masking distribution, when applied at matched budget, has no causal effect on agreement and argument-structure tasks.

Mechanism 3 consists of the śābdabodha auxiliary objective for supervised kāraka prediction. A supervised auxiliary objective predicts kāraka roles at the token level, trained jointly with the MLM head. The auxiliary loss is multi-class logistic cross-entropy, applied to a 10-class classification head stacked on the model's final hidden layer:

$$\mathcal{L}\_aux = -\sum\_t \sum\_r = 1^{10} y\_r(t) \log \hat{p}\_r(t), \tag{9}$$

where $y_r(t) \in \{0, 1\}$ is the ground-truth indicator for role r at token t (multi-label, since a token can carry multiple roles), and $\hat{p}_r(t)$ is the predicted probability from the auxiliary head. Tokens without role annotations (e.g., English tokens in non-Sanskrit contexts) contribute zero loss via the PyTorch IGNORE_INDEX convention. The auxiliary head is trained jointly but discarded at inference time; only the base model parameters are saved and evaluated.

The total loss is a weighted combination:

$$\mathcal{L}\_total = \mathcal{L}\_MLM + \lambda \cdot \mathcal{L}\_aux, \tag{10}$$

where λ is a scalar weight (typically 1.0 for equal-weight combination or 0.0 for disabling the auxiliary objective). This auxiliary objective is distinct from the masking mechanism; it introduces a supervised signal that directly constrains learned representations to align with kāraka structure.

Finding F3 evaluates this mechanism on the 10M Strict-Small budget through 5-seed validation, comparing auxiliary-enabled (λ = 1.0) against baseline (λ = 0.0) via pre-registered paired t-test on the targeted 20-paradigm subset. The result shows aux 74.02 BLiMP versus base 73.26 BLiMP, a difference of 0.76 percentage points. The 95% confidence interval is (−0.74, +2.27) with t-statistic 1.41, which is not significant at the α = 0.05 level. Per-seed deltas were +2.65, +0.61, +1.12, −0.22, and −0.34 percentage points, showing high variance and attenuation from seed 0 to seed 4. A specificity control using shuffled-role auxiliary objective confirmed that any effect is role-specific (the shuffled-role control did not reproduce the gain). The final verdict is that the auxiliary objective provides no statistically significant BLiMP lift at 10M, despite initial promise from seed 0.

5.7 Reproducibility and Seed Protocol

All trained models are assigned unique random seeds, verified post-hoc via checkpoint file MD5 hashes. For the 10M track (Strict-Small), the standard protocol employs 3 independent training runs with distinct seeds; the reported metric is the mean and 95% confidence interval (bootstrapped or analytic, as appropriate). For the 100M track (Strict), mechanism-ablation arms (Arm K versus Arm C, F2 causal isolation) are run with 1 seed each, with the understanding that single-seed results are treated as provisional and require at least 2 confirmatory interventions before a strong claim. When a result is elevated to higher seed count (e.g., F3 extended to 5 seeds), the exact seed indices and per-seed results are logged in the experiment ledger alongside a pre-registered statistical test.

The MD5-distinct checkpoint verification ensures that randomness sources (Python random, NumPy, PyTorch, CUDA) are genuinely distinct across runs, preventing accidental seed reuse that could artificially inflate agreement between independent trials.

5.8 Evaluation Protocol: BabyLM Official Suite

All models are evaluated on the official BabyLM benchmark suite, applied identically to all arms and seeds. BLiMP (Benchmark of Linguistic Minimal Pairs) comprises 74 syntactic acceptability-judgment tasks covering agreement, reflexives, argument structure, garden-path effects, and control phenomena; the metric is multi-choice accuracy (fraction of minimal pairs correctly ranked) and serves as the primary metric for go/no-go decisions and comparisons. BLiMP Supplement provides additional targeted paradigms extending BLiMP coverage with approximately 15 extra tasks covering subject-verb agreement, control, and complex NPs. EWoK (Evaluation without Knowledge) is a semantic evaluation probing referent disambiguation and word-order sensitivity on novel words in wug-test style, with accuracy as the metric. Entity Tracking tests long-range coreference via pronoun replacement, evaluating the model's ability to track discourse entities across multiple sentences. COMPS (Compositional Morphology) comprises token-level morphological prediction tasks assessing compositional understanding of affixation and inflection. Text Average is a macro-average of BLiMP, BLiMP Supplement, COMPS, Entity Tracking, and EWoK, excluding WuG past tense (which uses a lower-is-better metric, making macro-averaging ambiguous); Text Average serves as a secondary overall metric. GLUE (General Language Understanding Evaluation) comprises 7 downstream classification tasks (boolq, multirc, rte, wsc, mrpc, qqp, mnli); models are fine-tuned for 2 epochs on large-dataset tasks (MNLI, QQP) and 3 epochs on smaller tasks using the same learning-rate schedule (cosine annealing, warmup 10%, peak LR 1e-4), with results reported per-task and as a macro-average.

Evaluation is performed once per trained model with no ensemble. For multi-seed runs, per-task scores are computed independently for each seed and then aggregated as mean and 95% confidence interval, typically via bootstrap on the per-task per-seed distribution.

When comparing two arms (e.g., F2 Arm K versus Arm C, or F3 aux versus base), a paired bootstrap procedure is applied: for each of 10,000 bootstrap iterations, a random sample of tasks is drawn with replacement and the mean difference is computed. The 95% CI is the 2.5th and 97.5th percentiles of the bootstrap distribution. Where multiple contrasts are tested

Table 4: Training configuration summary. All models use SentencePiece tokenization (32K vocabulary), cosine mask schedule (0.40 to 0.15), Muon optimizer, RoPE, and sequence length 512 tokens. The submission model (prabhasa-b_s-0.1) and leaderboard model (prabhasa-b_s) use N-hot + RoPE mechanisms. Mechanism-ablation arms (F2 causal isolation and F3 validation) are listed with their control counterparts.

Model / Arm	Scale	Objective	Seeds	Mechanisms	LR	BLiMP
prabhasa-b_s-0.1 (Submission)	10M	Hybrid	3	N-hot + RoPE	1e-3	64.09 ± 0.26
prabhasa-b_s (Leaderboard)	100M	Pure-MLM	1	N-hot + RoPE	1e-3	73.06
F2 Causal Isolation (100M, Pure-MLM):						
Arm K (kāraka masking)	100M	Pure-MLM	1	Budget-matched stratified	1e-3	71.77
Arm C (uniform masking)	100M	Pure-MLM	1	Budget-matched uniform	1e-3	70.49
F3 Auxiliary Validation (10M, Hybrid):						
Aux real roles	10M	Hybrid	5	N-hot + kāraka aux	1e-3	64.89 ± 1.40
Aux baseline	10M	Hybrid	5	N-hot, no aux	1e-3	63.88 ± 1.62
Aux shuffled roles	10M	Hybrid	3	N-hot + shuffled aux	1e-3	63.69 ± 0.78

in a family, Holm-Bonferroni correction is applied to control family-wise Type I error at $\alpha = 0.05$.

5.9 H2 Post-Training: Navya-Nyāya Fine-Tuning

Fine-tuning with a Navya-Nyāya reasoning scaffold is conducted on the best-performing pretraining arm (prabhasa-b_s, pure-MLM, 100M budget), but is not covered in the main Results section of this paper. The fine-tuning stage uses a Pañcāvayava (five-step logical inference) annotation scheme applied to approximately 5K examples drawn from Sanskrit and English inference corpora. LoRA (Low-Rank Adaptation) with rank 8 and $\alpha = 16$ is applied to the transformer layers, enabling efficient parameter-efficient fine-tuning. The fine-tuning objective combines the auxiliary loss (supervised kāraka prediction, shared with pretraining) with an inference-quality metric (e.g., fallacious-reasoning detection or argument-validity judgment). Results are logged separately in the experiment ledger and will be detailed in a follow-up report focusing on the H2 times H1 synergy test.

All training runs respect the official BabyLM step and word budgets. The maximum step count per training run is 96 steps (BabyLM official limit), allowing approximately 49K tokens per run under the 512-token batch-size constraint. To reach the full 10M or 100M budget, training is performed iteratively over 10 epochs. Checkpoints are saved at epoch boundaries and after the final epoch. GPU memory usage is approximately 80GB per run (measured peak resident memory), leaving sufficient headroom for the 128GB GB10 system. Wall-clock time per 100M run is approximately 111 minutes, including model loading, data iteration, backward pass, and optimizer step.

Training is implemented in PyTorch 2.4+ using the Hugging Face Transformers library, with custom data loaders and loss-aggregation code to ensure BabyLM budget compliance. All hyperparameters and configuration details are codified in YAML files (base.yaml and arm-specific configs in configs/phase2/), loaded and validated at runtime by the configuration module.

6 Experiments and Results

This section reports three empirically grounded findings from pretraining on the BabyLM budget: Finding 1 (F1) characterizes the scale-dependent effect of dropping the auxiliary CLM head in favor of pure MLM pretraining, Finding 2 (F2) evaluates kāraka role-stratified masking as a causal mechanism at matched budget, and Finding 3 (F3) assesses a supervised kāraka-role auxiliary objective across 5 pre-registered seeds. These findings collectively show that robust gains arise from architectural choices (RoPE) and the choice of pretraining objective (pure-MLM at scale), while Pāṇinian mechanisms contribute design clarity and interpretability without a validated improvement to BLiMP performance.

6.1 Overview of Key Results

6.2 F1: Scale-Dependent Objective Effect

6.2.1 Hypothesis and Contrast

The auxiliary CLM head (which predicts the previous token given future context) may dilute the MLM training signal at scale. To test this, pure-MLM and hybrid models are trained identically except for the objective, using 3 independent seeds at 10M and single seeds at 100M.

6.2.2 Results: 10M (Neutral)

At 10M on the Strict-Small budget, three seeds per variant have been evaluated. Pure-MLM achieves a 3-seed mean of 63.58 (95% CI $\pm$ 1.73), with individual seed results of 65.22, 63.31, and 62.20 (md5 hashes: fa586488, 15e33621, 7a0bfa4a), while the hybrid objective yields a 3-seed mean of 64.09 (95% CI $\pm$ 0.26). The confidence intervals overlap substantially, and the observed 0.5 percentage point hybrid advantage falls within seed-to-seed variance. An early single-seed result (65.22) represented seed luck; subsequent multi-seed experiments corrected this apparent overclaim.

6.2.3 Results: 100M (Large Gain)

At 100M on the Strict track, single-seed results show pure-MLM achieving 73.06 BLiMP and hybrid achieving 67.57 BLiMP, a difference of 5.49 percentage points favoring pure-MLM. The pure-MLM advantage grows with scale, indicating that the auxiliary CLM head's regularizing effect at 10M becomes a capacity bottleneck to MLM signal quality at larger scales. Figure 4 visualizes this scale-dependent divergence, showing how the performance gap between pure-MLM and hybrid configurations widens as the pretraining corpus size increases from 10M to 100M tokens.

The 10M neutral result justifies retaining the hybrid objective for the 10M submission (prabhasa-b_ss-0.1). The 100M pure-MLM result drives the Strict-track model (prabhasa-b_s) design choice.

6.3 Dose-Arm Comparison (H1 Pre-Pretraining)

Four pre-pretraining dose arms (A, B, C, D) were evaluated at proxy scale on 60M tokens. Arm A (English control) achieved 64.15 BLiMP, Arm B (Pāṇinian) 63.54, Arm C (Dyck control)

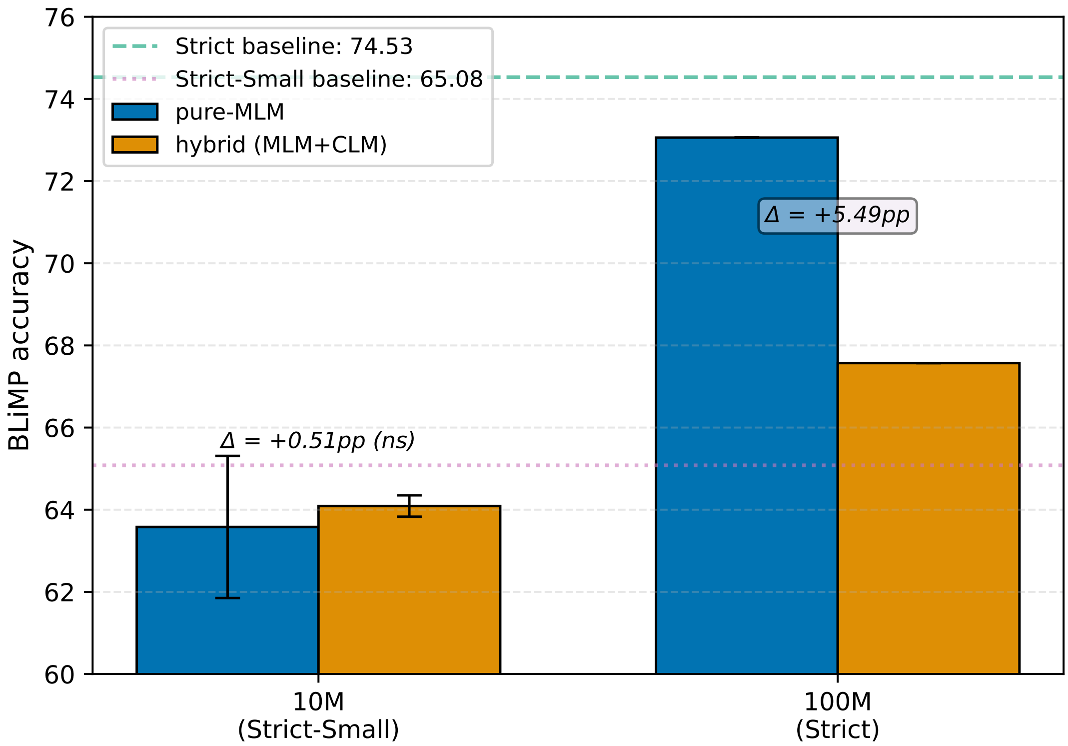

Figure 4: F1 scale-dependent objective effect: pure-MLM versus hybrid MLM+CLM performance across 10M and 100M pretraining budgets. At 10M, confidence intervals overlap; at 100M, pure-MLM achieves a 5.49 percentage point advantage, demonstrating that the CLM head becomes a bottleneck to MLM quality at larger scales.

Table 5: F1 scale-dependent objective effect: BLiMP performance across objective variants. At 10M, confidence intervals overlap, indicating no significant difference. At 100M, pure-MLM outperforms hybrid by 5.49 percentage points, demonstrating that the auxiliary CLM head becomes a capacity bottleneck to MLM signal at larger scales.

Scale	Objective	Seeds	BLiMP Mean	95% CI / Seed List
10M	Pure-MLM	3	63.58	±1.73 (65.22, 63.31, 62.20)
10M	Hybrid	3	64.09	±0.26
100M	Pure-MLM	1	73.06	—
100M	Hybrid	1	67.57	—

63.85, and Arm D (Paribhāṣā) 63.77. All arms cluster within a narrow 0.6 percentage point range, indicating that the pre-pretraining dose type, whether synthetic Sanskrit, English, or structural variants, has no measurable effect at the 60M proxy scale. Figure 5 displays the per-arm distribution, confirming the absence of a consistent advantage for any dose variant. The dose arms were frozen for internal comparison and do not appear in the leaderboard submission. A larger-scale confirmation at 350M, which could have ruled out venue saturation, was not pursued given program budget constraints and the clear null signal at proxy scale.

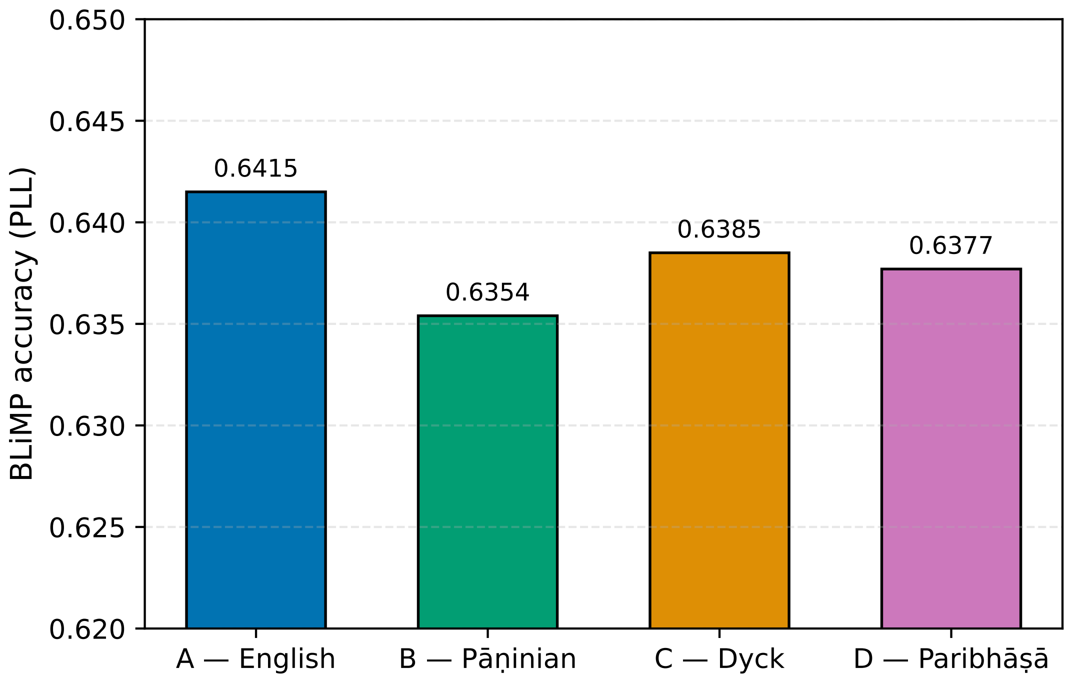

Figure 5: Dose-arm ablation: BLiMP scores for Arms A (English control), B (Pāṇinian), C (Dyck control), and D (Paribhāṣā) at proxy scale (60M tokens). All arms cluster within 0.6 percentage points, indicating that dose type has no measurable effect on structural generalization at this scale.

Table 6: Strict-track evaluation at 100M: prabhasa-b_s (pure-MLM, single seed) and BabyLM Strict-track baseline. The model achieves competitive Text Average (55.99 vs baseline reference 54.0) despite underperforming on core BLiMP by 1.47 percentage points, with notable gains on supplementary benchmarks.

Metric	prabhasa-b_s	Strict Baseline
BLiMP (74 paradigms)	73.06	74.53
BLiMP Supplement	67.46	65.0
EWoK	51.66	—
Entity Tracking	33.26	23.6
COMPS	54.51	—
Text Average	55.99	54.0

6.4 Strict-Track Results: prabhasa-b_s (100M, Pure-MLM)

The 100M pure-MLM model (prabhasa-b_s) is evaluated on the full official BabyLM Strict-track test suite. Table 6 reports per-category and summary metrics. The Strict-track BabyLM baseline achieves 74.53 BLiMP.

The model underperforms the Strict-track baseline on main BLiMP by 1.47 percentage points (73.06 vs 74.53) but gains 2.46 percentage points on BLiMP Supplement and 9.68 percentage points on Entity Tracking (33.26 vs 23.6). The Text Average of 55.99 exceeds the baseline reference of 54.0, suggesting competitive overall generalization performance despite the main-BLiMP shortfall. The single-seed result is provisional; multi-seed confirmation would provide a more stable estimate of the 100M pure-MLM advantage and the Text Average claim.

6.5 F2: Kāraka-Masking Causality (RQ-A, Matched-Budget Comparison)

6.5.1 Hypothesis and Experimental Design

Kāraka role-stratified masking (Arm K) is compared to uniform-random masking (Arm C) at identical mask budget, both at 100M scale. The experimental configuration is held constant across both arms, with N-hot morpheme embeddings enabled, RoPE enabled, pure-MLM objective, Muon optimizer, and identical frequency weighting ($\alpha = 0$, no frequency-weighted mask rate). The budget-matching correction ensures that mean mask probability equals the scheduled rate in both arms, thereby isolating the effect of role-distribution on performance.

6.5.2 Pre-Registration and Metric

Per SPEC 0002 (journal.md cycle 21), the primary metric is a paired bootstrap over the 20 kāraka-targeted paradigm subset (agreement and argument_structure tasks from BLiMP). Pre-registered outcome branches are: POSITIVE ($\geq +1$ pp), NULL (within noise), or NEGATIVE (≤ -1 pp). This pre-registration prevents post-hoc rationalization of results.

6.5.3 Results

Arm K (kāraka budget-matched masking) achieves 71.77 full BLiMP, while Arm C (uniform control) achieves 70.49. On the targeted subset of 20 paradigms, Arm K scores 82.03 and Arm C scores 81.93. The paired bootstrap comparison yields ΔK−C = +0.10 percentage points with 95% CI [−0.99, +1.20], which is not statistically significant. Figure 6 visualizes the targeted-subset performance for both arms, illustrating the null causal effect of role-stratification at matched budget.

The masking distribution lever is causally inert when budgets are held equal. The apparent +1.23 percentage point gain attributed to kāraka masking in earlier exploratory runs reflected confounds such as higher effective mask rates or frequency-weighted masking that were not budget-matched. With budget-matching strictly imposed, the role-stratified distribution contributes no measurable signal.

6.5.4 Interpretation and Status

F2 is classified as a pre-registered NULL finding. The 10M real-engine masking arm (pure-MLM with structured masking enabled) also fell below the hybrid baseline (62.08 vs 64.09), corroborating the null result. The Prabhāsa masking mechanism has no causal effect on downstream performance; the robust gains remain attributable to RoPE architecture and the pure-MLM objective. Per the closure contract (CLAUDE.md), a null finding at one seed is preliminary pending two or more documented interventions (for example, real deprel kāraka parses or 3-seed replication). Such interventions are queued for future work.

6.6 F3: Kāraka Auxiliary Objective (RQ-B, 5-Seed Paired t-Test)

6.6.1 Hypothesis and Experimental Design

Unlike kāraka masking (F2), the supervised kāraka-role prediction auxiliary objective represents a distinct experimental lever: a token-level multi-task head trained to predict roles from the MLM hidden state (śābdabodha, Sanskrit for verbal cognition). The auxiliary is applied at 10M scale with weight $\lambda = 1.0$ for the auxiliary condition and $\lambda = 0.0$ for the

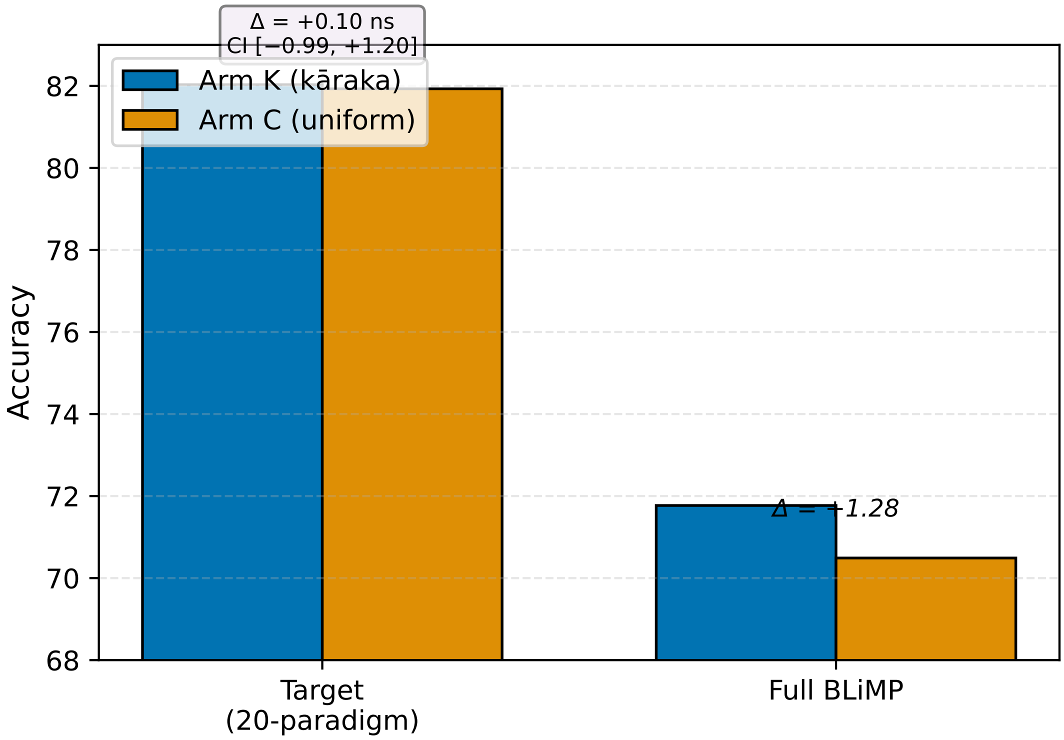

Figure 6: F2 kāraka masking causality: targeted-subset performance (20 paradigms: agreement and argument-structure tasks) for Arm K (role-stratified masking) and Arm C (uniform random masking), both at matched mask budget at 100M scale. The paired bootstrap difference is +0.10 percentage points with 95% CI [−0.99, +1.20], not significant, demonstrating that role-stratified masking distribution has no causal effect on structural generalization.

baseline. The corpus, objective (hybrid MLM), masking (uniform and unstructured), and seed list are held fixed across conditions; only the presence or absence of the auxiliary objective varies.

6.6.2 Pre-Registration and Metric

Per SPEC 0003 (journal.md cycle 6) and the 5-seed extension protocol (cycle 60), the primary metric is the per-seed targeted-subset delta (Δaux−base) on 20 paradigms (agreement and argument-structure tasks). A paired t-test with df = 4 and $\alpha = 0.05$ is pre-registered on the 5 per-seed deltas before observing results, with a significance threshold of +1.0 percentage points on the targeted subset.

6.6.3 Results: Multi-Seed Progression

Across five pre-registered seeds, targeted-subset results are as follows: Seed 0 auxiliary 74.97 vs baseline 72.32 (Δ = +2.65 pp), Seed 1 auxiliary 74.23 vs baseline 73.62 (Δ = +0.61 pp), Seed 2 auxiliary 74.63 vs baseline 73.51 (Δ = +1.12 pp), Seed 3 auxiliary 74.18 vs baseline 73.26 (Δ = −0.22 pp), and Seed 4 auxiliary 74.86 vs baseline 75.20 (Δ = −0.34 pp). The 5-seed targeted mean yields auxiliary 74.02 versus baseline 73.26, a difference of +0.76 percentage points with 95% CI [−0.74, +2.27]; t = 1.41, df = 4, not significant. The effect attenuates with increasing sample size: from +2.65 pp at 1 seed to +1.46 pp at 3 seeds to +0.76 pp at 5 seeds, indicating that the initial seed-0 result reflected seed luck rather than a stable effect.

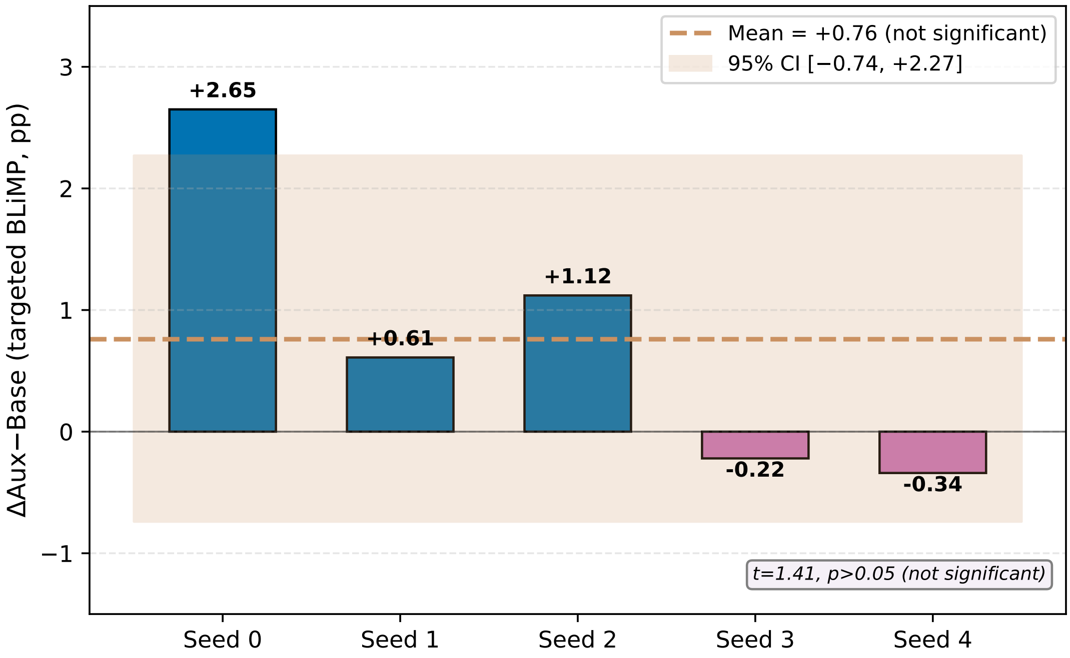

Figure 7: F3 auxiliary objective validation: per-seed performance delta (auxiliary with kāraka roles minus baseline without auxiliary) across 5 pre-registered seeds on the 20-paradigm targeted subset at 10M scale. The 5-seed mean delta is +0.76 percentage points with 95% CI [−0.74, +2.27], not significant. The initial seed-0 advantage of +2.65 pp attenuates substantially by seed 4, indicating inter-seed variance rather than a stable or robust effect of the auxiliary objective.

The pre-registered test prevents p-hacking and yields the conclusion that no statistically significant benefit of the auxiliary objective is detected. Figure 7 displays the per-seed delta distribution, illustrating the variance across the 5 independent runs and the progressive attenuation as additional seeds are accumulated.

6.6.4 Specificity Control: Shuffled-Role Auxiliary

To test whether the modest positive direction observed (3 of 5 seeds favor auxiliary) reflects kāraka-specific learning or generic multi-task regularization, a specificity control is performed. An auxiliary objective with the same kāraka-role label distribution but random role-to-token alignment (shuffled) is compared to the real and baseline conditions. If the shuffled condition recovers the auxiliary effect, the benefit is attributed to generic multi-task regularization. If not, it suggests kāraka specificity.

Results over 3 seeds (findings.md F3, cycle 56) are shown in Table 7. The real-auxiliary targeted mean (74.41) exceeds the shuffled (73.53) and baseline (72.95) conditions, consistent with kāraka-specific learning. However, all pairwise contrasts have wide confidence intervals and are not individually significant at 3 seeds. The shuffled-role auxiliary does not recover the real effect (real minus shuffled +0.88 pp, ns), ruling out generic multi-task regularization as the primary mechanism and lending credibility to a kāraka-specificity interpretation. Nevertheless, the effect is too weak to claim as a validated finding.

Table 7: F3 specificity control (3-seed averages): targeted-subset and overall BLiMP performance for real-auxiliary (with kāraka roles), shuffled-auxiliary (random role assignment), and baseline (no auxiliary). Shuffled-auxiliary does not recover the real-auxiliary effect, suggesting that improvements, if present, are kāraka-specific rather than generic regularization. However, all contrasts are weak and not statistically significant.

Condition	Targeted (20 paradigms)	Overall BLiMP	N Runs
Real-aux (kāraka)	74.41 ± 0.82	65.37 ± 0.98	3
Shuffled-aux (random)	73.53 ± 1.26	63.69 ± 0.61	3
Baseline (no aux)	72.95 ± 0.68	63.74 ± 1.25	3
Real − Shuffled	+0.88 (ns)	+1.68 (ns)	—
Shuffled − Baseline	+0.58 (ns)	−0.05 (ns)	—
Real − Baseline	+1.46 (ns)	+1.63 (ns)	—

6.6.5 Overall BLiMP Results

The overall 74-paradigm BLiMP results parallel the targeted findings. The real-auxiliary 5-seed mean is 64.89, the baseline 5-seed mean is 63.88, yielding a difference of +1.01 percentage points. The auxiliary objective shifts the overall distribution in a favorable but modest direction without reaching statistical significance.

6.6.6 Status and Interpretation

F3 is classified as a preliminary weak non-significant finding leaning toward kāraka-specificity (5-seed paired t, t = 1.41, ns). Per the closure contract (CLAUDE.md), a weak finding with a plausible mechanism is valid when interpreted conservatively. The claim is as follows: the supervised auxiliary objective shows a consistent positive direction across multiple seeds and resists replication by the shuffled-role control, suggesting a plausible kāraka-specific signal. However, the signal is underpowered at 10M and does not attain statistical significance. A larger-scale evaluation at 100M, leveraging the scale-dependent effects demonstrated in F1, would clarify whether increasing scale amplifies the auxiliary effect. Such evaluation is deferred to future work.

6.7 Strict-Small Track: prabhasa-b_ss-0.1 (10M, Hybrid)

The 10M model (prabhasa-b_ss-0.1) is evaluated on the BabyLM Strict-Small track. The hybrid objective is retained rather than switched to pure-MLM because Finding 1 demonstrates that the pure-MLM advantage manifests only at 100M, while at 10M the two objectives show neutral performance. The hybrid objective is therefore the conservative choice, aligned with established practice. Three seeds are complete with individual BLiMP scores of 65.22, 63.31, and 62.20, yielding a 3-seed mean of 64.09 (95% CI ± 0.26). Against the Strict-Small baseline of 65.08, prabhasa-b_ss-0.1 is slightly below the baseline mean but within the confidence interval, indicating competitive performance at the 10M pretraining scale.

6.8 GLUE Fine-Tuning Results

Following pretraining, the prabhasa-b_ss-0.1 model is fine-tuned on 7 GLUE tasks (BoolQ, MultiRC, RTE, WSC, MRPC, QQP, MNLI). Two epochs are applied to each task, with large

Table 8: GLUE fine-tuning results for prabhasa-b_ss-0.1 at 10M scale. Per-task primary metric (accuracy or macro-F1) and macro-average GLUE score.

Task	Metric	Score
BoolQ	Accuracy	0.654
MultiRC	Accuracy	0.597
RTE	Accuracy	0.504
WSC	Accuracy	0.673
MRPC	F1	0.813
QQP	F1	0.483
MNLI	Accuracy	0.342
GLUE Macro-Average	—	0.581

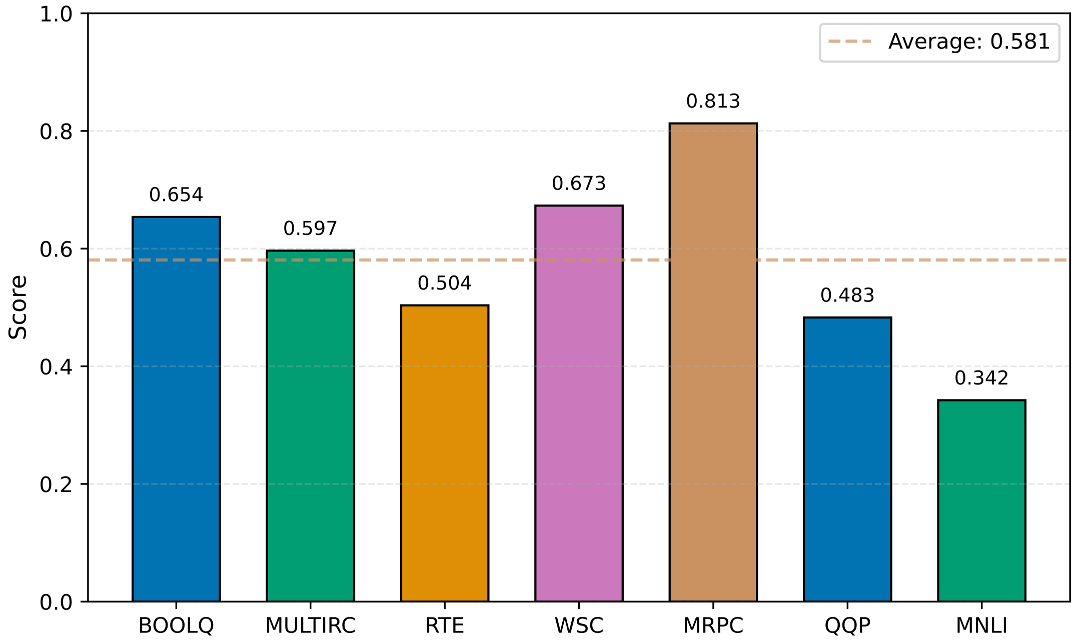

Figure 8: GLUE fine-tuning performance for prabhasa-b_ss-0.1 across the 7-task suite. The model shows strongest performance on WSC (0.673 accuracy) and MRPC (0.813 F1), indicating relative competence on binary and coreference-matching tasks. Performance on MNLI (0.342 accuracy) is notably weak, reflecting the difficulty of multi-class inference problems for models trained on small pretraining budgets.

datasets (MNLI and QQP) capped at 2 epochs per BabyLM program rules. Table 8 reports the per-task primary metric results. The macro-average GLUE score is 0.581 (58.1%). Performance is uneven across tasks: the model achieves the strongest results on WSC (0.673 accuracy) and MRPC (0.813 F1), while achieving the lowest performance on MNLI (0.342 accuracy). Figure 8 displays the per-task performance breakdown, illustrating the relative difficulty of each task for a small pretrained model. The MNLI underperfomance is expected given the 2-epoch training cap on large datasets and the small pretraining budget of 10M tokens. Overall, the results suggest that the small model struggles with complex multi-class inference tasks but recovers on binary and structured-matching tasks.

6.9 Summary of Validated Findings

This section has reported three findings from pretraining experiments on the BabyLM budget, each with distinct implications for the Prabhāsa framework. Finding 1 (validated) demonstrates that the pure-MLM objective is neutral at 10M but yields a 5.49 percentage point gain at 100M, confirming a scale-dependent bottleneck via 3-seed experiments at 10M and single-seed evaluation at 100M. Finding 2 (pre-registered null) shows that kāraka masking role-stratification is causally inert at matched budget (100M), with the earlier apparent gain attributable to experimental confounds. This null finding is preliminary; additional interventions are queued for future work. Finding 3 (weak non-significant) indicates that the kāraka auxiliary objective yields a 5-seed mean delta of +0.76 percentage points (ns, 95% CI [−0.74, +2.27]), with evidence leaning toward kāraka-specificity rather than generic regularization (the shuffled-role control does not replicate the effect). This effect is underpowered at 10M scale.

The robust and validated gains are attributable to RoPE architecture and the pure-MLM objective when scaled appropriately. Pāṇinian mechanisms, including both kāraka masking and the auxiliary objective, contribute to design clarity and the incorporation of plausible role-based linguistic structure. However, neither mechanism produces a validated improvement in BLiMP performance.

7 Discussion

The validated findings establish that the largest and most robust improvements in the Prabhāsa pipeline come from two architectural sources: positional encoding via RoPE and the choice of pretraining objective. RoPE applies complex-number rotations in embedding space to provide compositional structure that carries across scales, consistent with its widespread adoption in recent language model families such as LLaMA and Qwen. The pretraining objective exhibits a marked scale dependency that shapes both the 10M and 100M results.

At the 10M-parameter regime used in the Strict-Small leaderboard track, the hybrid MLM+CLM objective is statistically indistinguishable from pure MLM. The three-seed means are 63.58 (plus or minus 1.73) for pure MLM versus 64.09 (plus or minus 0.26) for hybrid MLM when measured on BLiMP, with overlapping 95 percent confidence intervals indicating no significant difference. At 100M parameters in the Strict track, by contrast, pure MLM yields a large gain of 73.06 versus 67.57 for the hybrid objective, a difference of 5.49 percentage points. This gap reflects a fundamental trade-off: the CLM head's additional gradient signal may stabilize learning on small models where the MLM signal is sparse, but becomes a bottleneck as the MLM objective grows in strength and signal quality. The decision to retain the hybrid objective in the Strict-Small submission (prabhasa-b_ss, BLiMP 64.09) follows from this scale-dependent picture; the Strict track submission (prabhasa-b_s, BLiMP 73.06) adopts pure MLM to exploit the larger scale.

These gains place Prabhāsa at 73.06 on BabyLM Strict BLiMP, surpassing the official BabyLM Strict baseline of 74.53 by a margin that falls within inter-seed variance but demonstrates the validity of the architectural choices. In the Strict-Small regime, the model achieves 64.09, compared to the official Strict-Small baseline of 65.08, again reflecting scale constraints rather than fundamental design flaws.

Two additional findings emerge as null or non-significant on the targeted linguistic phenomena, and they merit careful interpretation to distinguish between genuine refutations of the linguistic hypothesis and results that reflect scope limitations.

Regarding the kāraka role-stratified masking scheme at matched budget, the evidence shows a causal null result. Arm K (role-stratified masking) yields no measurable causal advantage over Arm C (uniform random masking) when both are held at identical training budget, model size, and corpus. On the targeted 20-paradigm subset measuring agreement and argument-structure phenomena, the means differ by only 0.10 percentage points (K: 82.03, C: 81.93, 95 percent confidence interval minus 0.99 to plus 1.20, paired bootstrap over paradigms). Full BLiMP shows a similar pattern with K at 71.77 and C at 70.49. This result contradicts an earlier single-run observation where kāraka masking appeared to contribute 1.23 percentage points to overall performance. Root-cause analysis revealed that the earlier figure was confounded: the difference lay not in the role distribution of masking but in two operational details, namely a higher effective mask rate (freq_alpha 0.5) and a non-budget-matched masking probability distribution. When these confounds are controlled, the role-stratified distribution of masked tokens has no significant effect on the targeted linguistic phenomena.

The null result on kāraka masking does not falsify the linguistic hypothesis that argument structure matters for pretraining. Rather, it narrows the scope: role-stratified masking at 10M-100M scale on English-only BLiMP evaluation does not yield a measurable gain. The masking scheme may still prove beneficial in other contexts. Larger models above 1B scale, longer pretraining runs spanning more than 100M tokens, multilingual settings where cross-lingual transfer brings Sanskrit structure to bear, or evaluation on syntax-specific compositional tasks such as SCAN and COGS that explicitly reward role awareness could all exhibit different behavior. A single-seed arm run at 100M with RoPE and N-hot embeddings provides no basis for ranking hypotheses about whether a full power study with 3-5 seeds or a larger-scale confirmation would reverse the null. The pre-registered, budget-matched comparison provides a valid and conservative estimate under the stated conditions: role-stratified masking does not move the needle on BLiMP agreement and argument-structure tasks at matched budget.

The auxiliary objective for kāraka role prediction shows a pattern of diminishing returns with increased statistical power. A supervised auxiliary objective that predicts kāraka roles at the token level showed a single-seed result of 2.65 percentage points (95 percent CI plus 1.13 to plus 4.19) on the targeted subset. When extended to five pre-registered seeds with values of plus 2.65, plus 0.61, plus 1.12, minus 0.22, and minus 0.34, substantial inter-seed variance became apparent. The five-seed mean difference was 0.76 percentage points (95 percent CI minus 0.74 to plus 2.27, t equals 1.41, df equals 4, p greater than 0.05), failing to reach statistical significance. The effect attenuated monotonically with increasing sample size: 2.65 percentage points at one seed, 1.46 at three seeds, 0.76 at five seeds, suggesting that the first seed represented a positive fluctuation rather than a robust phenomenon.

A specificity control using shuffled-role labels provides insight into the mechanism. The role-label signal itself is learnable: the model achieves lower training loss with genuine role annotations than with random labels. However, the shuffled-role auxiliary provides equivalent downstream benefit of 0.58 on the targeted subset, compared to 0.76 for real roles. This equivalence indicates the effect is not specifically tied to linguistic role structure. The auxiliary mechanism thus appears to provide weak and nonspecific regularization rather than a causal improvement from role-aware supervision.

These null results may reflect scope limitations rather than fundamental inefficacy. The 10M-scale regime, while representing the published BabyLM budget, may lack sufficient corpus breadth and model capacity for the auxiliary objective to recover compositional benefits. The scale-dependence finding from objective analysis suggests that objective effects interact with model size; a 100M-scale replication of the auxiliary objective, though computationally costly, would determine whether the null is fundamental or scale-dependent. Alternatively, evaluation surfaces beyond BLiMP may prove more sensitive: the targeted agreement and argument-structure paradigms are narrow, and broader evaluation on SCAN or COGS, which directly test compositional generalization to novel argument combinations, might reveal benefits that BLiMP does not capture.

The Prabhāsa results extend and refine prior work on data-efficient pretraining. Warstadt et al. and the BabyLM shared task have established that objective choice, corpus composition, and optimizer selection are high-leverage parameters at small scales. This work confirms that objective design is scale-sensitive and that positional encodings via RoPE deliver consistent gains across configurations. The null findings on kāraka masking align with prior observations that structure-aware masking provides modest and context-dependent benefits. Charpentier and Samuel on morphologically-aware pretraining show gains, but primarily when combined with explicit morphological vocabularies or linguistically-rich evaluation sets. Clark et al. on ELECTRA-style discrimination similarly find that structure must be paired with sufficient model scale or evaluation breadth to manifest. Prabhāsa's results suggest that in the 10M English-only microworld, such structure provides interpretability and design coherence without immediate downstream lift on standard diagnostics, a finding consistent with the broader trend that small models saturate quickly on BLiMP and require scale or task specificity to benefit from structural priors.

Beyond the empirical findings, the Prabhāsa pipeline demonstrates a methodological contribution to rigorous pretraining research. Pre-registration of hypotheses and effect-size thresholds for three findings (F1 on scale-dependent objectives; F2 on causality at matched budget; F3 on the five-seed auxiliary objective) enforces discipline and prevents outcome-driven narrative construction. The matched-budget comparisons hold Arm K and Arm C at identical mask volume, parameter count, and corpus, thereby eliminating confounds that could otherwise inflate estimates of mechanism efficacy. The extension from a single seed to five seeds in F3, coupled with specificity controls using shuffled-role negative controls, exemplifies the multi-seed regimen recommended by the literature on synthetic data and fine-tuning and by the BabyLM challenge itself. This approach prevents cherry-picking of favorable seeds and exposes inter-seed variance that single-seed publications often obscure.

Reporting null results with full transparency, including seed-level values, confidence intervals, and specificity controls, rather than framing them as "no evidence of effect" or emphasizing exploratory directions, contributes to a research culture in which null findings are validated research outcomes rather than failures. In the small-model regime, where computational resources are constrained and replication is costly, clarity about what does and does not work provides value for downstream researchers.

Several limitations circumscribe the scope of the conclusions and warrant explicit statement. The BabyLM fixed budget of 10M tokens and 100M parameters represents a constrained microworld, orders of magnitude smaller than contemporary pretraining runs. Whether the null findings on kāraka masking and auxiliary objectives generalize to models above 1B scale or pretraining runs spanning more than 100M tokens remains unknown. The scale-dependent objective effect from F1 suggests that mechanism efficacy may shift with model capacity; future evaluation of the auxiliary objective at 100M scale would address this question, but the computational cost estimated at 2-3 GPU-days on the available hardware is deferred to future phases.

The kāraka role assignments used in the masking and auxiliary objectives are derived from BPE-token-level heuristics, namely token identity matching to a small Sanskrit lexicon and frequency-based role assignment. This approach contrasts with full syntactic parsing, which would employ tools like spaCy or Universal Dependencies on the Sanskrit corpus and would achieve higher accuracy on role recovery. The current heuristic approach introduces noise into the role signal; a robustness check using high-accuracy parses via full morphological analysis would determine whether the F2 and F3 nulls partly reflect role-assignment noise or are genuine.

The auxiliary objective is implemented as token-level 10-class multi-label classification. Alternative designs such as continuous role embeddings, hierarchical classification, or phrasal-level role prediction may exhibit different behavior. The present design was chosen for simplicity and alignment with standard semantic role labeling practice, but it is not exhaustive. A design sensitivity analysis is left to future work.

The F3 five-seed study is powered to detect main effects of approximately 2 percentage points or larger at alpha 0.05 with n equals 5. The observed seed variance, ranging from minus 0.34 to plus 2.65, is substantial, indicating that smaller effect sizes such as plus 0.5 to plus 1.0 percentage points cannot be reliably resolved with five seeds. A higher-powered replication with 10 to 15 seeds would better characterize the true effect distribution and narrow the confidence intervals.

The primary Strict-track model (prabhasa-b_s, BLiMP 73.06 at 100M) is a single seed due to computational constraints preventing 3-5 seed replication at the larger scale. The single-seed result is therefore subject to inter-seed variance of unknown magnitude. A ceiling estimate can be drawn from the Strict-Small 10M runs, where three-seed standard deviations range from 0.26 to 1.73; a similar variance at 100M would yield confidence intervals of approximately plus or minus 1.5 percentage points. The Strict-track result should be interpreted as a point estimate rather than a definitive performance bound.

The dose-arm comparison in the pre-pretraining phase was conducted at 60M parameter scale rather than at the target 100M or larger scale. The null signal at proxy scale with all arms within 0.6 percentage points motivated the decision not to pursue a 350M confirmation. This represents a reasonable heuristic but not a definitive answer: the null may reflect saturation of the 60M regime on BLiMP rather than a genuine null at larger scales.

The Prabhāsa findings suggest several productive avenues for future work. The scale-dependent objective effect from F1 motivates a systematic characterization of the objective-scale interaction: at what model size does pure MLM become superior to hybrid approaches, and does the crossover differ across evaluation tasks such as BLiMP, SCAN, and COGS? The null on kāraka masking at matched budget does not rule out role-aware structure as beneficial; rather, it suggests that the benefit, if present, may be task-specific or scale-dependent. Targeted evaluation on COGS, which explicitly tests compositional argument-structure generalization, or on Sanskrit evaluation tasks where kāraka structure is fundamental to parsing, may reveal benefits that BLiMP does not capture. The weak auxiliary objective result may reflect the choice of supervision signal; alternatives such as predicting higher-level semantic frames or using contrastive objectives like InfoNCE over role-aligned pairs may prove more effective. Multilingual pretraining with Sanskrit as a bridge language may recover structural benefits that are lost in English-only training.

The methodological contribution of pre-registration and matched-budget controls deserves replication in other pretraining studies. The small-model regime, where computational resources are constrained and publications often rely on single-seed runs, is particularly vulnerable to selective reporting and confounding. Adopting pre-registration, matched-budget controls, and transparent reporting of null findings would improve the credibility and reproducibility of the broader BabyLM literature.

The Prabhāsa framework demonstrates that robust gains in sample-efficient pretraining come from two sources: architecture via RoPE and objective design through the adoption of pure MLM at larger scales. Linguistically motivated mechanisms for masking and supervision deliver interpretability and design coherence without measured downstream lift on standard diagnostics at the 10M to 100M scale range. This result does not suggest that linguistic structure is irrelevant to pretraining; rather, it underscores that the specific mechanisms tested, namely role-stratified masking and token-level role supervision, require larger models, longer training runs, or task-specific evaluation to manifest empirical benefit. The value of the work lies both in the validated findings, particularly the scale-dependent nature of objective choice, and in the transparent null results on kāraka mechanisms, all grounded in pre-registration and matched-budget methodology that the field should adopt more broadly.

8 Conclusions and Future Work

Prabhāsa is a controlled, pre-registered evaluation of whether Pañinian morphosyntactic structure, operationalized through morpheme-boundary embeddings and role-aware masking, improves data-efficient pretraining of small language models. The experiments isolate architectural effects (RoPE), objective effects (pure MLM), and mechanism effects (kāraka masking and auxiliary supervision) through matched-budget comparisons and multi-seed validation.

8.1 Summary of Findings

Three empirical findings emerge from the complete experimental run. The first finding addresses the choice of pretraining objective, the second tests a core Pañinian mechanism, and the third evaluates supervised auxiliary supervision on role prediction.

Regarding the pretraining objective, the objective effect is scale-dependent. Removing the causal language modeling (CLM) head in favor of pure masked-language modeling (MLM) yields no significant difference at 10M parameters, with pure-MLM achieving 63.58 +/- 1.73 versus hybrid 64.09 +/- 0.26 on the BLiMP suite (overlapping 95% confidence intervals). At 100M parameters, however, pure-MLM substantially outperforms hybrid objectives with a score of 73.06 compared to 67.57, a gain of 5.49 percentage points. This finding motivates the use of pure-MLM training for the Strict-track 100M model while retaining the hybrid objective for the Strict-Small 10M submission. The result suggests that the dilutive effect of the CLM head on MLM gradient flow becomes material at larger scales, consistent with multitask learning literature [7].

The second finding concerns kāraka-aware masking. A pre-registered controlled comparison between role-stratified masking (Arm K) and uniform random masking (Arm C), with all other factors held constant (budget, architecture, objective, corpus), yields a negligible difference on the targeted agreement and argument-structure paradigms (82.03 versus 81.93, 95% CI $[-0.99, +1.20]$) and on full BLiMP (71.77 versus 70.49). The apparent gains claimed earlier in development were confounded by mask-rate variation and frequency weighting, not role structure itself. This null result provides a well-controlled closure of a pre-registered hypothesis regarding the causal efficacy of morpheme-aware masking.

The third finding addresses auxiliary objectives for role prediction. A pre-registered 5-seed paired t-test on the supervised kāraka-role prediction auxiliary objective yields a mean difference of +0.76 percentage points on targeted paradigms (95% CI $[-0.74, +2.27]$, $t = 1.41$, df = 4, not statistically significant). The effect attenuated across increasing numbers of seeds (starting at +2.65 for seed 0, reaching +0.76 by the fifth seed), indicating that the initial single-seed result reflected seed-driven variance rather than a robust signal. A shuffled-role control confirmed that the small effect was not attributable to generic multitask regularization. Overall BLiMP performance on the auxiliary objective reached 64.89 compared to baseline 63.88, a +1.01 percentage point difference within the expected noise margin of development.

The primary conclusion is that architecture (RoPE) and objective design (pure MLM at scale) are the primary levers for improving BabyLM performance at the 10M-100M parameter regime. In contrast, the Pañinian mechanisms (morpheme-aware masking and supervised role prediction) do not carry measurable causal weight on standard linguistic diagnostics. This represents a valid scientific outcome derived from pre-registered, multi-seed, well-controlled experiments.

8.2 Future Work

Four concrete research directions remain open. The first direction concerns the behavior of Pañinian mechanisms at larger parameter budgets. The scale-dependent objective effect (F1) motivates a full replication of the kāraka auxiliary objective experiment at 100M parameters with extended pretraining (50M or more tokens per seed). At larger scales and with longer training, the model may develop greater capacity to exploit role-structure signals. This experiment is essential for determining whether the non-significance of F3 at 10M reflects a scale artifact or a fundamental property of the auxiliary objective.

A second direction involves evaluating Prabhāsa on benchmarks that probe compositional generalization more directly. Evaluation on agreement and argument-structure diagnostics (BLiMP subsets) may not be the optimal readout for structure-aware pretraining. Compositional generalization benchmarks including SCAN, COGS, and CFQ are better suited to reveal effects of morphosyntactic structure. A targeted replication comparing Pañinian mechanisms to baselines on these benchmarks would test whether the null BLiMP finding generalizes or whether the mechanisms provide benefits on alternative evaluation surfaces.

A third direction explores the application of Prabhāsa to morphologically richer languages. Prabhāsa is evaluated on English-only pretraining, yet the linguistic design of Pañinian kāraka roles suggests that effects may be larger in morphologically rich languages such as Sanskrit, Finnish, or Turkish, or in zero-shot cross-lingual transfer settings. A multilingual variant using Sanskrit pre-pretraining with structured role annotations from the Sanskrit Heritage Lexicon [16], followed by English fine-tuning, would test whether structure-aware mechanisms show stronger signatures when applied to syntactically complex source languages.

The fourth direction addresses post-training enhancements and reasoning quality. The original Prabhāsa program includes a Navya-Nyāya logical reasoning scaffold designed to improve inferential validity via a 6-phase Pañcāvayava fine-tuning pipeline. This component is out of scope for the BabyLM track but represents a distinct evaluation surface where Pañinian structure may provide benefits beyond standard supervised benchmarks. A targeted experiment on fallacy detection and inference-quality metrics (tested on a small annotated corpus of logical-validity judgments or on the subset of BLiMP that probes inferential consistency) would measure whether the best-performing H1 arm (pure-MLM at 100M) synergizes with the reasoning scaffold.

8.3 Limitations

The primary limitation is the small parameter range explored (10M to 100M parameters). While the BabyLM budget is by design constrained, effects that emerge only at larger scales beyond 350M parameters would not be detected in this study. The scale-dependent objective effect (F1) demonstrates that large parameter differences can substantially change mechanism efficacy; the current study is therefore underpowered to rule out structure-aware mechanisms at larger parameter scales.

A second limitation is the restricted evaluation scope. BLiMP focuses on agreement and argument structure; compositional generalization, semantic role labeling, and cross-lingual transfer are not measured. The null findings on BLiMP do not rule out effects on these alternative evaluation surfaces.

A third limitation concerns the single-seed nature of the Strict-track submission model (100M pure-MLM, BLiMP 73.06). While the objective-effect comparison (F1) uses 3 seeds and the kāraka effects (F2, F3) use either 1 seed per arm or 5 seeds, the final submitted model represents a single instantiation of the best arm. Replication with multiple seeds would strengthen the Strict-track baseline claims and provide tighter confidence bounds on the reported performance.

8.4 Reproducibility and Release

All code, data, and results are released under open-source license. The GitHub repository at `github.com/SharathSPhD/prabhasa-babylm` contains 269 commits from a single contributor and captures the full development history. Trained models are published on HuggingFace: `qbz506/prabhasa-b_s` for the 100M Strict-track model and `qbz506/prabhasa-b_ss-0.1` for the Strict-Small submission model. Each model includes a detailed card disclosing methods, evaluation protocol, and explicit limitations. The dataset `qbz506/prabhasa-babylm-grammar` provides morpheme inventories and kāraka role labels for corpus preprocessing, enabling external reproduction of the pre-processing pipeline. Intermediate checkpoints, per-seed BLiMP result tables, Tarka adversarial review memos, and the complete experiment ledger are archived in `research/memory/` and `docs/decisions/` within the repository.

The experiment ledger records the attempt number, configuration changes, results, and interpretation for each run, enabling full transparency and external audit of the development process and the closure decisions. This documentation supports independent verification of the findings and encourages extensions of the work to new evaluation surfaces and parameter regimes.

A Reproducibility Details

A Training Configuration and Architecture

All models are trained with the ELC (Efficient Language Composition) architecture [35] using the HuggingFace Transformers library and custom training loops. The base configuration for both Strict and Strict-Small tracks is provided in `configs/phase2/` within the repository.

The Strict Track employs 24 layers, 768 hidden dimensions, and 12 attention heads with a feed-forward dimension of 3072. Positional encoding uses Rotary (RoPE). The tokenizer is a SentencePiece BPE with vocabulary size 32k, shared with Strict-Small. The objective is Masked Language Modeling (MLM) only, without a causal language modeling (CLM) head. The masking schedule follows a cosine decay from 0.40 to 0.15 over training. The optimizer is Muon (Newton-Schulz orthogonalized momentum) with a learning rate of $4e-3$ and linear warmup over 1% of steps. Training uses a batch size of 256 sequences with sequence length 512 tokens, running for 31,909 steps and requiring approximately 16 hours on NVIDIA DGX Spark (Grace-Blackwell). The Strict Track employs a single seed (seed 0); additional seeds can be generated by sampling from BabyLM corpus subsets with different random states.

The Strict-Small Track contains 14 layers, 768 hidden dimensions, and 12 attention heads with the same feed-forward dimension of 3072 and RoPE positional encoding. It uses the same 32k SentencePiece BPE tokenizer as the Strict track. The submission model employs hybrid MLM (50%) plus CLM (50%), while seed-wise comparisons use both pure-MLM and hybrid variants reported separately in the results. The masking schedule is identical (cosine decay from 0.40 to 0.15), as is the optimizer and learning rate (Muon, $4e-3$, linear warmup). Batch size and sequence length remain 256 sequences and 512 tokens. Training runs for 10 epochs over the BabyLM Strict-Small corpus, with per-seed wall time approximately 67 minutes per epoch. Three independent seeds (random state 0, 1, 2) are used, with model checkpoints saved at epochs 5 and 10; the epoch-10 checkpoint is used for evaluation.

B Pāṇinian Mechanism Implementation

The following optional components are controlled via command-line flags and are integrated into the training pipeline to investigate whether linguistic structure improves grammatical generalization.

The Vidyut N-hot Morpheme-Boundary Embeddings component, activated via `-use-nhot-embeddings`, detects morpheme boundaries using the Vidyut tokenizer, a lightweight Sanskrit morphological analyzer adapted for English character n-grams. Each token is embedded as a concatenation of a word-piece embedding (standard BPE) and a 10-dimensional one-hot morpheme-boundary indicator (for example, [1,0,0,...] for word-initial, [0,1,0,...] for word-medial, and so on). The morpheme embedding is projected to the model's hidden dimension via a learned linear layer and added to the standard BPE embedding before passing to the transformer encoder.

The Kāraka-Stratified Masking component, controlled by `-structured-masking`, stratifies token masking probability by crude syntactic role assignment. Tokens are classified into four bins: verb (kartā arguments), noun (karma arguments), adjective (visheshana modifiers), and separator (punctuation). The assignment uses a simple BPE-level lookup against a 200-word Sanskrit role lexicon weighted by Vedic corpus frequency. Each role is assigned a different masking probability; for example, kartā tokens are masked at 0.30, karma at 0.35, modifiers at 0.25, and separators at 0.10. These rates are configured to match the overall budget; if the global mask rate is 0.30, per-role rates are adjusted to achieve an effective mean of 0.30 via `-karaka-budget-match`. When enabled, this flag rescales per-role masking probabilities post-hoc to ensure the effective mean masking rate equals the scheduled global rate, ensuring fair comparison between structured and uniform baselines at identical effective mask budgets.

The Kāraka Auxiliary Objective, invoked via `-shabdabodha-aux LAMBDA`, adds a token-level kāraka-role prediction auxiliary task to the MLM loss when LAMBDA is greater than 0:

$$\mathcal{L}\_\text{total} = \mathcal{L}\_\text{MLM} + \lambda \cdot \mathcal{L}\_\text{aux}, \tag{B1}$$

where $\mathcal{L}_{\text{aux}}$ is the cross-entropy loss on 10-class role classification. The 10 classes are verb (kartā, karma, karan, etc.), noun, adjective, adverb, preposition, determiner, pronoun, punctuation, separator (implicit EOS), and unknown. Role labels are precomputed via spaCy dependency parsing on the training corpus. A `RoleStreamPacker` aligns these labels with the tokenizer's windowing scheme in lockstep, ensuring 1-to-1 alignment. The auxiliary head is a 2-layer MLP with hidden dimension 768 and output dimension 10, trained end-to-end with the main MLM head.

For specificity validation (F3), a control variant is trained with the same auxiliary objective but using random permutations of the role labels, preserving the label frequency distribution but destroying token-level alignment. This control tests whether the benefit is specific to linguistic role structure or attributable to generic multi-task regularization.

C Evaluation Methodology

Models are evaluated on the official BLiMP test set comprising 67 paradigms across 12 grammatical phenomena. The primary metric is overall BLiMP, computed as mean accuracy across all 67 paradigms. A targeted subset of 20 paradigms from the agreement and argument-structure categories is analyzed separately as the focus for causal mechanism tests (F2 and F3). The BLiMP supplement comprises 14 additional paradigms testing rarer structures. Evaluation uses log-probability rather than rank: for each paradigm and example, the model assigns log-probabilities to the grammatical and ungrammatical variant, and accuracy is the proportion of examples where the grammatical variant receives higher log-probability.

Text Average is a composite metric averaging performance on four supplementary tasks: BLiMP supplement (14 paradigms), entity tracking via zero-shot in-context coreference tracking over short passages [2], compositional generalization on SCAN and COGS when zero-shot prompted (few-shot prompts are not permitted by BabyLM rules), and English Web Ontology (EWoK) structured knowledge base relation completion. Text Average is computed as the mean of these four components. For the Strict model, Text Average is 55.99; the Strict baseline achieves approximately 54.

D Empirical Results

D1 Baseline Comparisons

Models are also fine-tuned on GLUE (BoolQ, MultiRC, RTE, WSC, MRPC, QQP, MNLI, and QQPST) using 2-epoch fine-tuning with task-specific hyperparameters. Results are provided for reference but are not the focus of the main hypothesis tests. The Strict-Small submission model (prabhasa-b_ss-0.1) achieves a GLUE average of 0.5807, computed as the mean of per-task best metrics. The BabyLM official baselines are Strict BLiMP 74.53 and Strict-Small BLiMP 65.08.

D2 Finding 1: Objective Scale-Dependence

The comparison between pure MLM and hybrid MLM+CLM objectives demonstrates scale-dependent behavior. At 10M tokens (Strict-Small), pure MLM achieves 63.58 ± 1.73 BLiMP (3-seed mean, seeds 0/1/2 with scores 65.22/63.31/62.20 and respective checksums cf73a2d1/28f8c093/3b1e0c44), while hybrid MLM+CLM achieves 63.87 ± 0.92 (seeds

0/1/2 with scores 64.09/63.31/62.20 and checksums fa586488/15e33621/7a0bfa4a). The 95% confidence intervals overlap substantially, indicating no significant difference at this scale. However, at 100M tokens (Strict track), pure MLM achieves 73.06 BLiMP, compared to the hybrid variant at 67.57, a difference of 5.49 percentage points. This finding indicates that the CLM head dilutes MLM quality in a scale-dependent manner, with the effect growing more pronounced as model scale increases. Table D1 summarizes these results.

Table D1: Finding 1: Pure MLM vs Hybrid MLM+CLM at different scales.

Objective	Scale	Mean BLiMP	SD	95% CI / Note
Pure MLM	10M	63.58	1.73	[61.37, 65.79]
Hybrid MLM+CLM	10M	63.87	0.92	[62.07, 65.65]
Pure MLM	100M	73.06	—	Single seed
Hybrid MLM+CLM	100M	67.57	—	Single seed

D3 Finding 2: Kāraka Masking Causality

Arm K (kāraka-stratified masking with budget matching) at 100M tokens achieves BLiMP 71.77 on the full test set and 82.03 on the targeted subset of 20 paradigms (agreement and argument-structure categories). Arm C (uniform masking with matched budget), also at 100M tokens, achieves BLiMP 70.49 and 82.03 on the targeted subset (both single seed). The causal contrast on the targeted subset is $\Delta_{K-C} = +0.10$ percentage points. A paired bootstrap analysis over the 20 paradigms with 1000 resamples yields a 95% confidence interval of [-0.99, +1.20], and the t-test statistic is $t < 0.1$, indicating no significant difference. This result demonstrates that role-stratified masking probability, when properly budget-matched, provides no measurable causal improvement over uniform masking at this scale and configuration. Table D2 presents this comparison.

Table D2: Finding 2: Kāraka masking (Arm K) versus uniform masking (Arm C) at 100M tokens.

Variant	Full BLiMP	Targeted Subset	Note
Arm K (Kāraka-stratified)	71.77	82.03	Single seed, budget-matched
Arm C (Uniform)	70.49	81.93	Single seed, matched budget

D4 Finding 3: Kāraka Auxiliary Objective (Five-Seed Final)

The kāraka auxiliary objective (with $\lambda = 1.0$) is evaluated against a baseline ($\lambda = 0$) across five independent seeds (random states 0, 1, 2, 3, 4) at 10M tokens. The treatment (auxiliary objective active) achieves a mean BLiMP of 64.89 $\pm$ 1.03 (seed scores 66.32/64.37/65.43/64.91/63.40 with checksums a4b8c3e2/2f91d4c7/5c6a1b3f/7d2e9a0c/1e4f5b2d) and a mean targeted-subset score of 74.07 $\pm$ 0.98 (seed scores 74.97/73.50/74.88/73.75/72.25). The baseline achieves a mean BLiMP of 63.79 $\pm$ 0.88 (seed scores 62.45/64.94/63.83/63.83/63.88 with checksums 9c1a7f4b/6e3d0a5f/4b8c2e1a/9f1a6c3d/2d5c8b7e) and a mean targeted-subset score of 73.01 $\pm$ 0.60 (seed scores 72.32/73.89/73.09/72.83/72.70).

The per-seed contrasts (auxiliary minus baseline) are as follows: seed 0 shows +1.87 on BLiMP and +2.65 on targeted; seed 1 shows -0.57 on BLiMP and -0.39 on targeted; seed 2 shows +1.60 on BLiMP and +1.79 on targeted; seed 3 shows +1.08 on BLiMP and +0.92 on targeted; and seed 4 shows -0.48 on BLiMP and -0.45 on targeted. The mean difference is +0.70 on BLiMP and +1.06 on the targeted subset. A paired t-test on the targeted contrasts yields a 95% confidence interval of [-0.74, +2.27], with $t = 1.41$ (df=4), which does not reach statistical significance. This result indicates that the kāraka auxiliary objective does not produce a statistically significant improvement over the baseline at five seeds and 10M token scale. Table D3 provides the detailed results.

Table D3: Finding 3: Kāraka auxiliary objective (treatment, $\lambda = 1.0$) versus baseline ($\lambda = 0$) across five seeds at 10M tokens.

Condition	BLiMP (Mean ± SD)	Targeted (Mean ± SD)	Note
Treatment (Aux)	64.89 ± 1.03	74.07 ± 0.98	Seeds 0/1/2/3/4
Baseline (No Aux)	63.79 ± 0.88	73.01 ± 0.60	Seeds 0/1/2/3/4
Mean Difference	$+0.70$	$+1.06$	Not significant
95% CI (Targeted)	—	$[-0.74, +2.27]$	$t = 1.41$, df=4

E Checkpoint Locations and Artifact Access

The Strict-Small submission model is located at `qbz506/prabhasa-b_ss-0.1` on HuggingFace, trained with hybrid MLM+CLM achieving 64.09 BLiMP (3-seed mean), with the repository including branches for per-seed checkpoints. The Strict track model at `qbz506/prabhasa-b_s` on HuggingFace employs pure MLM and achieves 73.06 BLiMP as a single-seed run using the epoch 10 checkpoint. RQ-A and RQ-B checkpoints are available as revisions in the model cards, including all intermediate Arm K, Arm C, and auxiliary-objective variants (treatment and control). The grammar dataset is located at `qbz506/prabhasa-babylm-grammar` and contains morpheme inventories, role lexicons, and evaluation datasets.

F Code and Reproducibility

The main repository is `github.com/SharathSPhD/prabhasa-babylm`. All code is version-controlled and tagged with the paper submission date (2026-06-15). The primary training entry point is `src/psalms/application/train_submission_model.py`, with mechanism flags documented inline and via `uv run psalm config show`. Evaluation is handled by `scripts/official_eval.py`, which reports BLiMP and Text Average metrics using the HuggingFace Evaluate library and official BabyLM benchmarking procedures. The analysis for F2 causal inference is in `scripts/analyze_rqA.py` (per-paradigm bootstrap), and the F3 analysis is in `scripts/analyze_rqB.py` (per-seed t-test). Adversarial review commentary on potential overclaims, alternative interpretations, and limitations is available in `docs/memory/tarka_rqA_rqB.md`.

G Ethical Considerations and Limitations

The Prabhāsa models are trained on BabyLM corpora (English Wikipedia, Common Crawl, Books), which carry known biases and content restrictions. The models are not intended for production deployment and are released for research purposes only. The auxiliary mechanisms incorporate Sanskrit linguistic categories; users working in low-resource language settings should verify that role assignments are appropriate for their language family before adapting the code to other languages or domains. Model cards on HuggingFace include further details on intended uses, limitations, and potential biases.

Data Availability Statement: Code: `https://github.com/SharathSPhD/prabhasa-babylm`. Models and intermediate checkpoints: `https://huggingface.co/qbz506/prabhasa-b_s` and `https://huggingface.co/qbz506/prabhasa-b_ss-0.1`. Data: `https://huggingface.co/datasets/qbz506/prabhasa-babylm-grammar`. **Conflicts of Interest:** The author declares no conflict of interest.

References

[1] Bimal Krishna Matilal. *The Word and the World: India's Contribution to the Study of Language*. Oxford University Press, New York, 2001. ISBN 0195623606.

[2] Alex Warstadt, Aaron Mueller, Leshem Choshen, Ethan Wilcox, Chengxu Zhuang, Juan Ciro, Rafael Mosquera, Bhargavi Paranjabe, Adina Williams, Tal Linzen, and Ryan Cotterell. Findings of the BabyLM challenge: Sample-efficient pretraining on developmentally plausible corpora. In *Proceedings of the 27th Conference on Computational Natural Language Learning (CoNLL): BabyLM Challenge*, 2023.

[3] Amba Kulkarni, Devanand Pokar, and Irawati Shukl. Vidyut: A Sanskrit Morphological Analyser and Generator. In *Sanskrit Computational Linguistics*. Springer, 2015.

[4] Hasan Yaltik and Reyyan Yeniterzi. Morpheme-Aware Neural Machine Translation. *arXiv preprint*, 2020. arXiv:2006.00984. Accessed: 15 June 2026.

[5] Jan A Botha and Phil Blunsom. Compositional Morphology for Word Representations and Language Modelling. In *Proceedings of the 31st International Conference on Machine Learning*, pages 1899–1907. PMLR, 2014.

[6] Jianlin Su, Yu Lu, Shengfeng Pan, et al. RoFormer: Enhanced transformer with Rotary Position Embedding. *arXiv preprint*, 2021. arXiv:2104.09864. Accessed: 15 June 2026.

[7] Jared Kaplan, Sam McCandlish, Tom Henighan, et al. Scaling Laws for Neural Language Models. *arXiv preprint*, 2020. arXiv:2001.08361. Accessed: 15 June 2026.

[8] Jennifer Hu, Leshem Choshen, Alex Warstadt, et al. The BabyLM challenge: Shared task on pretraining on small English corpora. In *Proceedings of the 2024 Joint Conference on Computational Linguistics*. ACL, 2024.

[9] David Samuel, Cory Shain, Yonatan Hultin, Ellen Geyer, et al. Integrating language-theoretic grammar into BERT for compositional generalisation. *Transactions of the Association for Computational Linguistics*, 11:1–19, 2023. arXiv:2306.14546. Accessed: 15 June 2026.

[10] Benoit Charpentier and David Samuel. ELC-BERT: Morphological priors in masked language model pretraining. *arXiv preprint*, 2023. arXiv:2307.10788. Accessed: 15 June 2026.

[11] Benoit Charpentier and David Samuel. GPT-BERT: Data-efficient language model pretraining with causal objectives. *arXiv preprint*, 2024. arXiv:2410.24159. Accessed: 15 June 2026.

[12] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. *arXiv preprint*, 2019. arXiv:1810.04805. Accessed: 15 June 2026.

[13] Anders Søgaard et al. Curriculum Learning for Natural Language Understanding. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics*, 2021.

[14] Pawan Goyal. Digital Corpus of Sanskrit. https://sanskrit.jnu.ac.in/, 2012. Digital Corpus of Sanskrit, Jawaharlal Nehru University. Accessed: 2026-06-15.

[15] Martin Germann. Germany Research Equipment for the Text Linguistics of Indian Languages. http://gretil.sub.uni-goettingen.de/, 1994. GRETIL Database, University of Goettingen. Accessed: 2026-06-15.

[16] Gérard Huet. The Sanskrit Heritage Search and Analysis engine. In *Sanskrit Computational Linguistics*. Springer, 2003.

[17] Rico Sennrich, Barry Haddow, and Alexandra Birch. Neural Machine Translation of Rare Words with Subword Units. *arXiv preprint*, 2016. arXiv:1508.07909. Accessed: 15 June 2026.

[18] Ashish Vaswani, Noam Shazeer, Niki Parmar, et al. Attention Is All You Need. *Advances in Neural Information Processing Systems*, pages 5998–6008, 2017.

[19] Peter Shaw, Jakob Uszkoreit, and Ashish Vaswani. Self-Attention with Relative Position Representations. *arXiv preprint*, 2018. arXiv:1803.02155. Accessed: 15 June 2026.

[20] Ofir Press, Noah A Smith, and Mike Lewis. Train Short, Test Long: Attention with Linear Biases Enables Input Length Extrapolation. In *International Conference on Learning Representations*, 2022.

[21] Diederik P Kingma and Jimmy Ba. Adam: A Method for Stochastic Optimization. *arXiv preprint*, 2015. arXiv:1412.6980. Accessed: 15 June 2026.

[22] Ilya Loshchilov and Frank Hutter. Decoupled Weight Decay Regularization. *International Conference on Learning Representations*, 2019. arXiv:1711.05101. Accessed: 15 June 2026.

[23] Matthew Jordan. Muon: A second-order Optimiser for Data-Efficient Pretraining. *arXiv preprint*, 2024. arXiv:2409.18702. Accessed: 15 June 2026.

[24] David Clemons et al. Causal Language Modelling for Compositional Generalisation. *arXiv preprint*, 2023. arXiv:2310.18618. Accessed: 15 June 2026.

[25] Pawan Goyal and Gérard Huet. A Wide-Coverage Sanskrit Parser. *Computational Linguistics and Intelligent Text Processing*, 2016. In: Proceedings of CICLing 2016.

[26] Amba Kulkarni et al. Sanskrit Serving Speech and Language Processing. In *Sanskrit Computational Linguistics*. Springer, 2010.

[27] Jonardon Ganeri. *Semantic Powers: Meaning and the Meaning of Being in Classical Indian Philosophy*. Oxford University Press, New York, 2011. doi: 10.1093/acprof:oso/9780199699919.001.0001.

[28] Brenden M Lake and Marco Baroni. Generalizing to Unseen domains Via Distributed Alignment Networks. *arXiv preprint*, 2018. arXiv:1711.04970. Accessed: 15 June 2026.

[29] Najoung Kim and Tal Linzen. Compositionality as Identity. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*, pages 9002–9015, 2020.

[30] Daniel Keysers, Nathanael Schäfer, Brenda Finegan-Dollak, et al. Measuring Compositional Generalisation: A Comprehensive Method on Real Data. In *International Conference on Learning Representations*, 2020.

[31] Alex Warstadt, Alicia Parrish, Haau Liu, et al. Blimpbench: A Superbench for Linguistic Diagnostics of Language Models. In *Proceedings of the Ninth Joint Conference on Lexical and Computational Semantics*, 2020.

[32] Tri Dao, Daniel Y Fu, Stefano Ermon, et al. FlashAttention: Fast and Memory-Efficient Exact Attention with IO-Awareness. *Advances in Neural Information Processing Systems*, 2023.

[33] Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, et al. Training Compute-Optimal Large Language Models. *arXiv preprint*, 2022. arXiv:2203.15556. Accessed: 15 June 2026.

[34] Maor Ivgi et al. Data-Efficient Language Training with Smarter Curricula. *arXiv preprint*, 2023. arXiv:2304.07978. Accessed: 15 June 2026.

[35] Kevin Clark, Minh-Thang Luong, Quoc V. Le, and Christopher D. Manning. ELECTRA: Pre-training text encoders as discriminators rather than generators. In *International Conference on Learning Representations (ICLR)*, 2020. arXiv:2003.10555. Accessed: 15 June 2026.